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Full Review Article

Advances in porous volumetric solar receivers and enhancement of volumetric absorption

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ABSTRACT

Porous volumetric solar receivers are one type of solar receivers that can volumetrically absorb solar radiation and achieve efficient solar-to-thermal energy conversion. Porous volumetric solar receivers have been developed since 1980s. In this review, we focus on the development progress of the atmospheric and pressurized porous volumetric solar receivers, in which the structural designs, the material selections, the experimental research methods, the comparison of thermal performance, and the transient response characteristic of the receivers were reviewed. On the other hand, the theoretical research methods including the direct pore-scale and volume averaging simulations were introduced. The pore-scale reconstruction method and the procedure to investigate the fluid flow and heat transfer processes at the pore-scale were presented. For the volume averaging method, detailed descriptions for the selection of empirical parameters in the governing equations to be solved were summarized. Typical research results based on these methods were presented and research limitations were also pointed out. Furthermore, the methods for the enhancement of volumetric absorption and the improvement of thermal efficiency of the receivers have been comprehensively reviewed. Two methods including geometrical parameters optimization and spectrally selective absorption were presented in detail. This review will provide a better understanding of the development and research methods for porous volumetric solar receivers, and inspire future studies for the performance improvement of the receivers.

1. Introduction

The large-scale application of clean, low-carbon renewable energies has become a global consensus to resolve pressing issues such as climate change, environmental pollution, and energy shortage. Among various types of renewable energies, solar energy has a bright prospect due to its vast amount, clean utilization and easy access. In order to overcome the major problems associated with solar energy harvesting such as low energy density and inefficient energy conversion, concentrating solar power (CSP) technology has become one of important methods to convert solar energy to electricity. Based on the CSP Gen3 demonstration roadmap, the operation temperature of the system should be further elevated to above 700 °C for realizing the targeted 50% solar-to-electricity conversion efficiency [1,2]. However, there still exist many challenges in the solar receiver design, high temperature material selection, energy storage subsystem optimization. In a CSP system, the solar receiver is directly irradiated by solar radiation and its safe and high-efficiency operation is the key to increase the overall solar-to-electricity conversion efficiency and reduce the levelized cost of energy (LCOE) [3,4].

Based on the solar absorption process, solar receivers could be mainly divided into three types: tubular receiver, volumetric receiver, and particle receiver. The tubular receiver absorbs solar radiation on its outer surface and conducted the thermal energy to the heat transfer fluid inside the tube. The low thermal conductivity of high temperature alloy and the low convective heat transfer coefficient between tube and heat transfer fluid create a large thermal resistance, which leads to high receiver temperature and large thermal radiative loss [5,6]. Despite using the fluid to absorb and transfer heat, the particle receiver converts and stores solar energy based on solid particles. Depending on the flow direction of the solid particles, the particle receiver could be further divided into downflow, upflow, and horizontal flow types. The light leakage, particle flow control, and corrosion issues should be examined to improve the performance of the particle receiver [7,8]. On the other hand, the volumetric solar receiver distributes the incoming solar radiation into a three

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Nomenclature

a	absorption coefficient
c_f	inertia coefficient
D_p	particle diameter (m)
d	diameter (m)
g	asymmetry factor
h	convective heat transfer coefficient (W/(m^2·K))
K	permeability (m^2)
Nu	Nusselt number
n	refractive index
Pr	Prandtl number
p	pressure (Pa)
q	Heat flux (W/m^2)
Re	Reynolds number
T	temperature (K)
U	superficial velocity (m/s)

Greek symbols

β	extinction coefficient (m^{-1})
ε	porosity
λ	thermal conductivity (W/(m·K))
μ	dynamic viscosity (Pa·s)
σ λ	Stefan-Boltzmann constant
σ_s	scattering coefficient

Subscripts

c	cell
d	diameter
eff	effective value
f	fluid
h	hydraulic
r	radiation
s	solid

dimensional space. The concentration of solar flux on the receiver surface is reduced and the maximum allowable solar flux on the aperture of the receiver is increased.

In general, the volumetric absorption of solar radiation could be implemented by both the heat transfer fluid and porous media. The commonly used heat transfer fluids have low extinction coefficients to solar radiation and nanoparticles made of metals, nonmetallic oxides, and carbon materials are usually introduced [9,10] to form nanofluids, as shown in Fig. 1(a). By changing the particle type, size, and concentration, the optical properties of the nanofluids could be adjusted and the solar radiation could be gradually absorbed. In order to overcome the stability issue of nanofluids, the concentrated solar power on demand (CSPonD) [11,12] proposed a collocated receiver and storage system based on the solar radiation absorption by molten salt as shown in Fig. 1(b). Due to the low extinction of molten salt, the entire depth of the molten salt tank should be over several meters.

On the other hand, porous materials are widely used for solar absorption due to its large porosity and interconnected porous skeleton. The common porous structures include honeycombs, wire meshes, packed beds, and foams made from metallic or ceramic materials. The solar absorption and conversion processes inside the porous media are illustrated in Fig. 1(c). The concentrated solar radiation penetrates into the porous structure and is gradually absorbed. The heat transfer fluid (normally air) passes through the porous structure and exchange heat with the high temperature porous skeleton. In the ideal case, the fluid and porous medium temperature increases gradually along the main flow direction, which means that the receiver surface temperature is greatly reduced and this phenomenon is known as a volumetric effect [16,17].

Porous volumetric solar receivers show potential for increasing the solar-to-thermal conversion efficiency, but the heat transfer characteristics such as strong coupling, multiscale, and multimode pose challenges in experimental and numerical analyses. Firstly, a set of parameters such as geometries, optical and thermophysical properties of the porous media as well as the incident solar radiation distribution simultaneously influence the fluid flow and heat transfer processes in porous volumetric solar receivers. Secondly, the small scale of porous channels and the large scale of CSP systems require multiscale modeling of fluid flow and heat transfer in the porous volumetric solar receiver. Thirdly, multiple heat transfer mechanisms such as solar radiation transfer, heat conduction in the porous media, convective heat transfer between the porous skeleton and the heat transfer fluid, and radiative heat transfer create difficulties in evaluating the performances of the porous volumetric solar receiver. Therefore, many efforts have been taken on the porous volumetric solar receiver to delve deeply into the heat transfer process and put forward new methods for enhancing volumetric absorption.

In this review, the development of porous volumetric solar receivers is elaborated and the progresses in experimental and numerical studies are thoroughly reviewed. The methods to enhance the volumetric absorption and improve the thermal performance of porous volumetric solar receivers are comprehensively summarized.

2. Advances in experiment research on porous volumetric solar receivers

The porous volumetric solar receiver could be categorized as atmospheric (open) and pressurized types, and the main difference is the operating pressure of the heat transfer fluid. The major component of these two types of receivers is a porous structure, which usually uses high temperature resistant materials such as nicked-based superalloy (Inconel 600, Inconel 601, Alloy 230) and refractory ceramics (SiSiC, reSiC). Based on the intrinsic absorption or coating material, the incident solar radiation can be efficiently and gradually absorbed inside the porous structure. The typical structures of atmospheric and pressurized porous volumetric solar receivers are presented in Fig. 2. Atmospheric porous volumetric solar receivers have the porous structure, supporting structure, and air channel. The porous structure is directly exposed to the

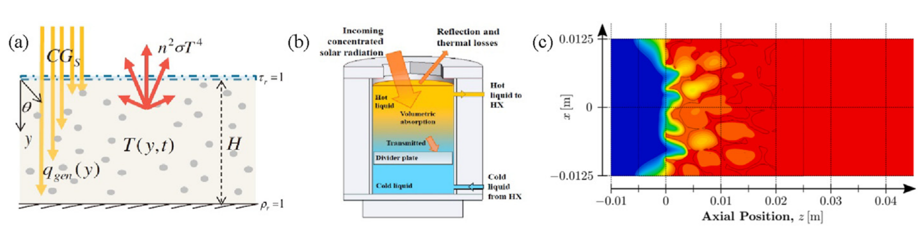

Fig. 1. Volumetric absorption of solar energy (a)Nanofluids (Reproduced with permission [13]. Copyright 2012, Elsevier); (b)Molten salts (Reproduced with permission [14]. Copyright 2018, Elsevier); (c)Porous media (Adapted with permission [15]. Copyright 2020, Elsevier).

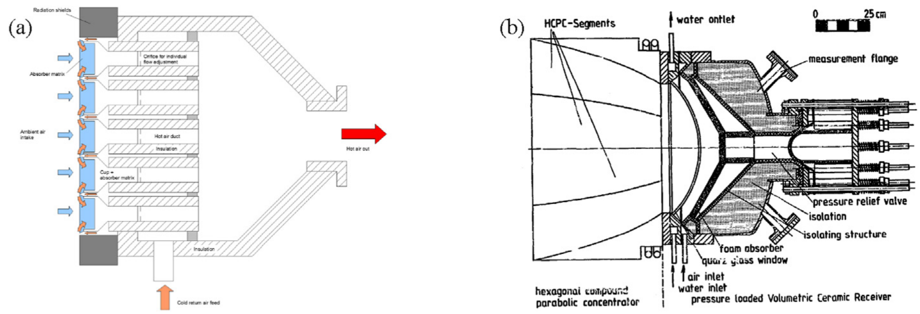

Fig. 2. Schematics of porous volumetric solar receivers (a)HiTRec type [18] Reproduced under the terms of the CC-BY licenses; (b) PLVCR500 (Reproduced with permission [19]. Copyright 1991, Elsevier).

incident solar radiation in atmosphere, and the ambient air is directly sucked into the receiver. In order to scale up the system and conveniently replace the damaged porous structure, the large-scale atmospheric porous volumetric solar receiver usually adopts a modular design.

As for the pressurized porous volumetric solar receiver, more complex structures need to be designed for maintaining the high operating pressure, which could reach up to 2 MPa. In general, a highly transparent window is necessary for sealing the cavity and allowing the solar radiation to pass through. It should endure high temperature, large thermal stress, and high pressure while having high transparency in the solar spectrum. For efficiently reducing the window temperature, the inlet air is usually directed to the window to cool it down and the preheated air is beneficial for increasing the overall thermal efficiency. The secondary concentrator is usually equipped to the pressurized porous volumetric solar receiver, which offers possibility to further increase the air outlet temperature to above 1000 °C. However, the poor design and limited machining precision may lead to non-uniform solar flux distribution and large thermal stress. For example, the poor adhesive between the mirror and background may cause breakage in the mirrors from overheating. Additional water-cooling subsystem is needed to reduce the high temperature of the secondary concentrator which increases the complexity and unreliability of the system.

2.1. Development progress in the porous volumetric solar receivers

Experimental research on porous volumetric solar receivers began in the 1980s. Fricker applied a wire mesh as the absorber and installed it on the solar dish [20]. The air passed through the wire mesh and reached 842 °C at the outlet of the receiver. Based on this solar thermal energy utilization method, the porous volumetric solar receiver has been extensively studied. Sulzer 1 and Sulzer 2 were the pilot receivers with output power over 200 kWth, which were both integrated into the CSP system at Plataforma Solar de Almeria (PSA) [20,21]. Despite the lower thermal efficiency compared with the estimated value, the Sulzer technology has accumulated experience in system operation and receiver transient performance testing. In 1993, Phoebus-TSA was tested at PSA which was the first megawatt-scale atmospheric porous volumetric solar receiver [22]. Its modular receiver was in the hexagon shape with a diameter of 280 mm and the entire aperture diameter of the receiver was 3400 mm. The outlet air temperature and the output power of the receiver reached 700 °C and 2.9 MW, respectively. The Phoebus-TSA also introduced the air-return system to reutilize the thermal energy of the exhaled air. The thermal efficiency reached 85% when the air return ratio and the outlet temperature were 60% and 700 °C, respectively. Moreover, the Phoebus-TSA attempted to integrate thermal storage, steam generator, and electric power generation subsystems. The steam temperature and pressure ranges were 480-540 °C and 3.5-15 MPa, respectively [20,23]. It is known that the outlet temperature of air is usually limited by the maximum allowable temperature of the porous media. Therefore, German Aerospace Center (DLR) promoted the application of ceramic materials for the porous volumetric solar receiver. In the same period of Sulzer, the CeramTec made of SiSiC was designed by DLR which had a staggered layout of the front. The reduced front surface decreased the reflective loss such that the maximum outlet air temperature reached 782 °C [24]. Hereafter, many ceramic atmospheric porous volumetric solar receivers were designed and tested, in which HiTRec and SOLAIR made this technology mature [25]. The honeycomb SiC porous structure was selected by HiTRec [26] and SOLAIR [27], and the modular design concept and air return system were adopted. The outlet air temperature of 800 °C was achieved with the thermal efficiency exceeding 80% for HiTRec I, HiTRec II, and SOLAIR 200. Compared with SOLAIR 200, the SOLAIR 3000 system further raised the number of modular receivers from 36 to 270, and the incident solar power reached 2950 kW [20,28]. In 2009, Solar Tower Jülich shown in Fig. 3 applied the technology of HiTRec and was in solar operation for a long period of time [29,30]. Under the solar only operation mode, the outlet air temperature achieved 650 °C and a 1 MW electric power was delivered to the regional grid.

On the other hand, coupling the CSP system with the combined cycle can improve solar energy utilization efficiency and reduce the LCOE, which promotes the development of the pressurized porous volumetric solar receiver. DLR developed PLVRCs (Pressure Loaded Volumetric Ceramic Receiver) with rated thermal power of 5 kW and 500 kW and put them into test in the solar furnace of Sandia and PSA, respectively [19]. The outlet air temperature of PLVRC-5 achieved 1000 °C while the air pressure was kept at 0.42 MPa. However, the performance of PLVRC-500 deviated largely from the design condition due to the insufficient cooling in the secondary concentrator and the non-uniform air mass flow distribution. REFOS [32,33] applied wire meshes as the absorber and improved the secondary concentrator design to improve the optical efficiency. During the test in PSA, the outlet air temperature achieved 800 °C but the cracks in the transparent window was also observed. Directly-Irradiated Annular Pressurized Receiver (DIAPR) [34] designed by Weizmann Institute of Science utilized a frustum-like high-pressure window, which improved the capability of withstanding high pressure (1.7-2.0 MPa). It arranged the absorbers with porcupine blocks, and the enhanced convective heat transfer led to a high outlet air temperature of 1200 °C with a thermal efficiency of 71%. Based on the technologies of REFOS and DIAPR, SOLGATE [35] and DIAPR multistage [20] connected the cavity receiver and the pressurized porous volumetric solar receiver for efficient utilization of solar radiation with different intensities. For example, SOLGATE arranged the low temperature receiver and the high temperature receiver in series and preheated the air in the low temperature receiver. The outlet air temperature achieved 800 °C, which generated electric power of 230 kW. Despite that Soltrec [36] and other studies [37,38] conducted at the lab-scale further investigated the performance of pressurized porous volumetric solar receivers, the development of this type of receivers was slow and no receiver at large scale was built. Considering for the high efficiency of combined cycle and the simple scheme of atmospheric porous volumetric solar receivers, the CAPTure project introduced regenerative heat exchangers and combined these two technologies [39]. The atmospheric heating and pressurized cooling processes were separated as shown in Fig. 4, and the

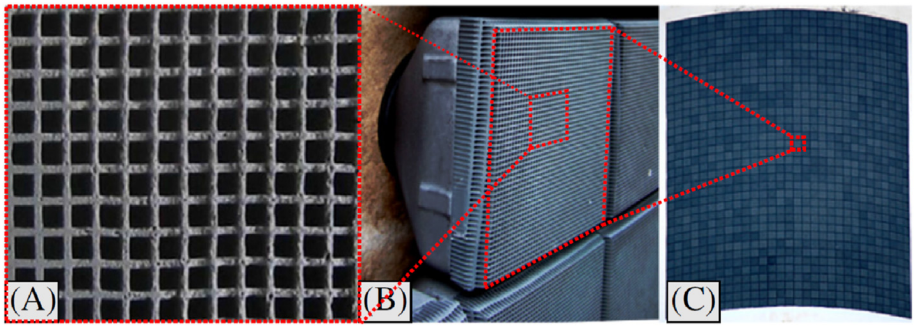

Fig. 3. Solar Tower Jülich volumetric solar receiver (Reproduced with permission [31]. Copyright 2017, Elsevier).

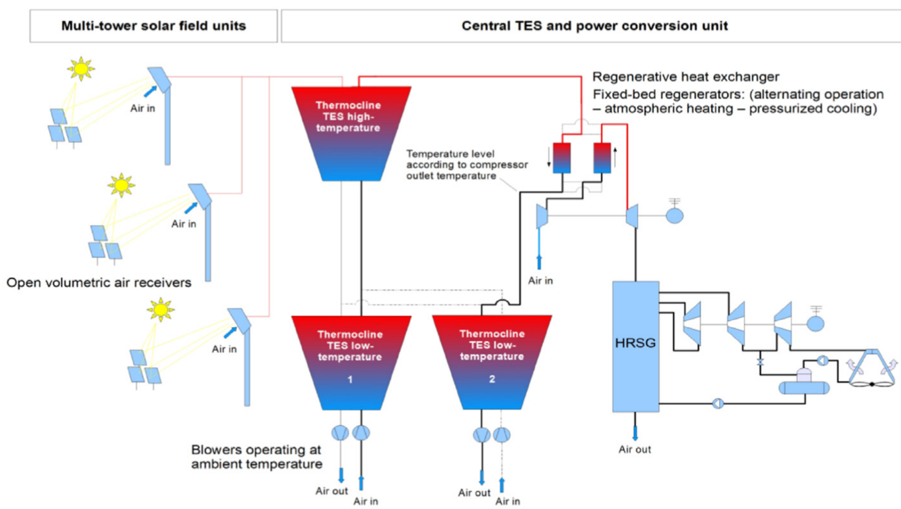

Fig. 4. CAPTure project solar combined cycle (Reproduced with permission [40]. Copyright 2022, AIP Publishing).

solar-to-electric efficiency of the solar combined cycle was 29.6% under 500 suns. Recently, Zaversky et al. [40,41] presented that the combined cycle introduced by CAPTure project was not economically competitive and in-field test encountered critical issues related to the turbine start-up. A hybrid CSP power plant coupled with compressed air energy storage was proposed which showed potential to increase the shares of renewables for the future power grid.

2.2. Thermal efficiency analysis of porous volumetric solar receivers

The outlet air temperature and the thermal efficiency are the two key indicators to evaluate solar receivers. The outlet air temperature is determined by many factors such as the intensity and distribution of incident solar radiation, the air mass flow rate and distribution, as well as the structure and geometrical parameters of the porous media. The higher the outlet air temperature is, the more efficient the power cycle is. The higher thermal efficiency of the receiver could be achieved by limiting the thermally conductive, convective and radiative losses. A higher thermal efficiency means that the solar radiation is effectively captured, absorbed and converted.

The thermal efficiency of a receiver is defined as the ratio of absorbed energy by the heat transfer fluid and the total incident solar energy. As for the absorbed energy by the receiver, the direct or indirect measurements could be used. The inlet and outlet air temperatures can be monitored by the thermocouples, and the absorbed energy can be calculated by temperature increase and air mass flow rate. In the experiments, the non-uniform distribution of solar radiation usually leads to a non-uniform distribution of air temperature. Therefore, more thermocouples are needed for measuring the air temperature distribution. On the other hand, the indirect method based on the energy conservation could also be applied to evaluated the thermal efficiency. Despite monitoring the fluid temperature, the indirect method measures the absorbed energy by the heat transfer fluid by applying a heat exchanger to cool down the air, which can be evaluated based on the temperature variation and efficiency of the heat exchanger. Ideally, the absorbed energy by the heat transfer fluid can be all removed by the heat exchanger. One advantage of the indirect method is that the measurement of air mass flow rate is no longer needed.

The incident solar radiation distribution could also be measured by direct and indirect methods. The direct method applied heat flux gage to measure the solar flux in the aperture of the receiver. In order to capture the solar flux map, multiple gages arranged regularly are usually needed. In these studies [21,24,42,43], multiple heat flux gages were placed in a cross shape and the solar flux in the two perpendicular directions is captured. This measurement presents a small deviation as the solar flux distribution is centrosymmetric. In HiTRec II, a special heat flux gage with a quick response time was adopted and installed on a mobile bar [26]. The mobile bar swept through the receiver aperture and quickly recorded solar flux data at different positions. Apparently, this quick response heat flux gage offered a higher spatial resolution but required delicate system design and components coordination. At the lab scale, the diameter of solar flux is relatively small and the heat flux gage could be precisely moved by the mapping system [44]. The indirect method uses the Lambertian target and CCD camera to capture the grey level image of solar radiation. The intensity of solar radiation is linked to the grey level of the image which is calibrated by the heat flux gage at one fixed position, such as HERMES and ProHERMES [45]. In this method, the Lambertian target can be a large plate which covers the entire receiver aperture or a small mobile plate which cooperates with the camera for continuous shooting. For the large scale solar receiver where the Lambertian target is not easy to be placed, the surface of the solar receiver can be used as the target as presented in the Solar Tower Jülich system [29]. Compared with the direct method, the indirect method offers a high spatial resolution and quick measurement. But the calibration of the measurement systems has become a major difficulty due to the non-linear relationship between the grey level image and the heat flux map as well as the spectral reflectivity of the target.

Fig. 5 summarized the experimental tests results including the outlet air temperature and the thermal efficiency of porous volumetric solar receivers since 1980s. The porous volumetric solar receiver was categorized into three groups as follows: receiver with no air return system or preheating, receiver with air return system, and receiver with preheating. In general, the atmospheric porous volumetric solar receiver with air return system can increase the air outlet temperature while giving a higher thermal efficiency. In HiTRec, SOLAIR, and TSA, the air return ratio was controlled in the range of 40%-60%. It was estimated that the thermal efficiency could be improved by 8% when the air return ratio increases from 60% to 80% when maintaining the air outlet temperature at 650 °C [31]. Most of the receivers with preheating were the pressurized type, and only Bechtel atmospheric porous volumetric solar receiver was designed to have a Pyrex globe for preheating the inlet air. Under different experimental cases, the temperature increases of air due to the preheating could account for 20%-30% of the total air temperature increase [43]. With regards to the pressurized porous volumetric solar receiver, the outlet air temperature was generally higher due to the concentrated solar radiation and air preheating. The outlet air temperature of the pressurized porous volumetric solar receiver PLVCR-5 and DIAPR exceeded 1000 °C. The maximum outlet air temperature for the atmospheric porous volumetric solar receiver was achieved by Mey-Cloutier et al. [46], in which the incident solar radiation was homogenized by a homogenizer and the outlet air temperature reached 958 °C. Moreover, it is seen from Fig. 5 that the ranges of the outlet air temperature and thermal efficiency are large, so it is difficult to directly compare two receivers under different test conditions. Despite that some studies [26,27] recommended to use air outlet temperature-mass specific energy graphic for comparing the receivers, it is still hard to evaluate different receivers under different test conditions (e.g. solar radiation intensity, distribution, air mass flow).

2.3. Transient performance of porous volumetric solar receivers

The porous volumetric solar receiver has a quick transient response time and small thermal inertia due to the shorter flow path of the heat transfer fluid. Up to now, there is no standard for the receiver transient performance and many studies adopted time constant to restabilize the system after changing the solar radiation intensity. In the transient test of Sulzer [21], the number of mirrors in the heliostat was reduced when the outlet air temperature was stable. The incident solar energy was reduced by 20%, and the outlet air temperature was correspondingly decreased. By monitoring the variation of the outlet air temperature, it was found that it took 6 min to restabilize the system and the time constant was about 90 s. With a similar test method, the time constants for Sandia foam [47], CeramTec [24], and HiTRec II [26] are 365 s, 660 s, and 70 s, respectively. The parametric and non-parametric methods were both adopted for SOLAIR [28] and the time constant of the receiver was determined to 600-840 s. In the test of DIAPR [34], the receiver regained 90% of the temperature drop within less than 5 min after reopening the shutter. Besides, some studies used the temperature increasing rate to evaluate the transient performance of the porous volumetric solar receiver. The temperature increasing rate of PLVCR-5 [19] was measured which achieved 80 °C/s. Zhu et al. [37] measured the outlet air temperature increase process and found that time to stabilize the system was about 30 min.

From the above studies, the time constant for porous volumetric solar receivers is usually in the range of tens of seconds to hundreds of seconds. Comparing the metallic and ceramic receivers, the transient response time is normally smaller for the metallic receiver because of the higher thermal conductivity. The quicker transient response time of porous volumetric solar receivers is beneficial of fast start-up, but it needs higher requirements in maintaining the stability of the outlet air temperature and formulating system control strategies.

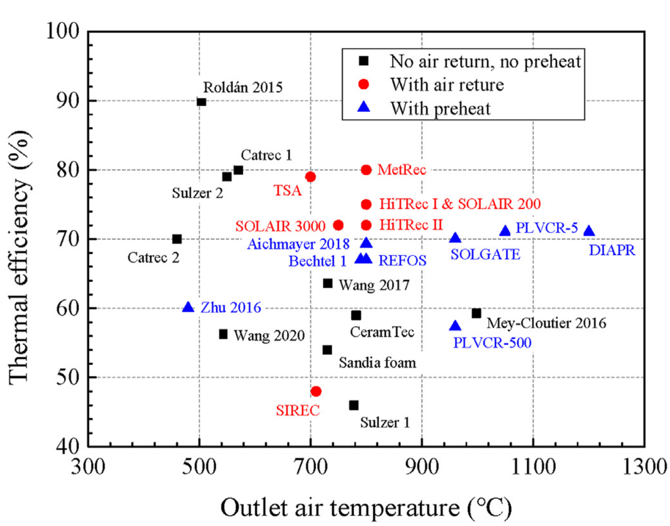

Fig. 5. Thermal efficiency of porous volumetric solar receivers.

3. Advances in theoretical research on porous volumetric solar receivers

To investigate the detailed fluid flow and heat transfer processes, the multiscale simulation methods are applied. When the sophisticated porous structures are considered, the simulation studies are performed at pore-scale. The primary task is to reconstruct the porous geometry based on the periodic unit cell, random, or X-ray CT reconstruction methods. Then the solar radiation transfer and coupled heat transfer could be solved based on direct pore-scale simulation method. On the other hand, the porous domain can be treated at Representative Elementary Volume (REV) scale. The influences of porous skeleton to the fluid flow and heat transfer of the fluid are introduced by empirical parameters. Generally, the direct pore-scale numerical simulation gives a precise result but requires more computing resource. The volume-averaging simulation method provide a quick performance evaluation, but its accuracy is largely dependent on the proper determination of empirical parameters.

3.1. Direct pore-scale numerical simulation method

Direct pore-scale numerical simulation methods investigate the fluid flow and heat transfer processes based on the detailed reconstructed models of the porous media. It is an effective simulation method to reveal the transport process inside the porous media and to determine the empirical parameters. The reconstruction of the porous media and the establishment of the heat transfer model for porous volumetric solar receivers are presented as follows.

3.1.1. Pore-scale model reconstruction of the porous media

The porous media with different structures have been widely used as porous volumetric solar receivers, such as honeycombs, wire meshes, and foams. In order to enlarge the penetration depth of solar radiation, the porous media usually have large porosity and interconnected channels. Therefore, the pore-scale model of high porosity porous media is presented and the reconstruction method for the low porosity porous media such as packed beds and rocks is not the focus of this study. Based on the structure of the porous media, the pore-scale reconstruction method can be divided into periodic unit cell reconstruction method, random reconstruction method, and X-ray CT reconstruction method.

3.1.1.1. Periodic unit cell reconstruction method. Since many porous media have certain periodic characteristics, the periodic unit cell reconstruction method simplifies the porous skeleton into a spatial array of ideal structural units. For example, honeycombs and wire meshes have strict periodicities as shown in Fig. 6(a) and (b). The periodic unit structure can be easily extracted and the model reconstruction of these types of porous media are not introduced. For the porous foam, it shows some randomness as illustrated in Fig. 6(c), but it does have some periodicities in analogy with aggregation of soap bubbles. The parameters such as unit cell diameter, pore diameter, number of vertexes and edges can be used to describe the porous structures.

Some studies [53,54] adopted a simple cube as the unit cell as shown in Fig. 7(a). It simplifies the porous foam to a large extent, ignoring the arrangement of the porous skeleton in other directions. By adjusting the cell size and skeleton thickness, this model could control the pore density and porosity of the reconstructed porous media. Moreover, the processing characteristics during the fabrication of the porous media and more structural features are included in the reconstruction process. For graphite porous media, it has an approximately circular pore structure, thus Yu et al. [55] and Karimian et al. [56] used a sphere-centered cube as the unit cell as shown in Fig. 7(b). Similarly, the cell diameter and pore diameter can be adjusted by changing the edge size and sphere diameter in the unit cell. In the foaming process, the formation of bubble aggregates is usually controlled by surface tension which leads to a lower surface energy of each bubble. Therefore, the Kelvin structure has been widely applied for the reconstruction of porous foams as illustrated in Fig. 7(c). The Kelvin structure is formed by the truncated octahedron with 6 square faces and 8 hexagonal faces, which has been considered to be the unit body to equally divide the space with the smallest surface area. Krishnan et al. [57,58] subtracted the cube with a sphere and got a Kelvin structure with round pores in the faces. Bai and Chung [59] subtracted the truncated octahedron with a sphere, and a similar Kelvin structure was obtained. Compared with the Kelvin structure of Krishnan et al., the Kelvin structure of Bai and Chung had straight edges. It is noticed that the porous skeleton constructed by the above two methods is thicker near the vertex and smaller at the center, showing the characteristics of the skeleton thickness change caused during fabrication process. In 1994, Weaire and Phelan [60] proposed a unit body that divides the space better than the Kelvin structure and further minimizes the surface area by 0.3%. As shown in Fig. 7(d), it is composed of six 14-sided polyhedra and two 12-sided polyhedra. Boomsma [61] applied the software of Surface Evolver to gradually minimize the surface energy of the Weaire-Phelan structure and obtained the porous skeleton with triangular cross section.

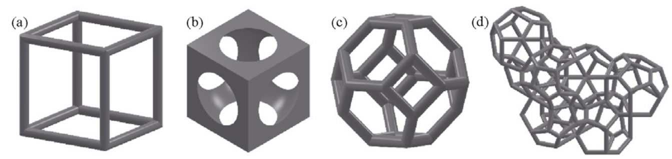

Fig. 7. Periodic structures of porous media (a)Simple cube; (b) Sphere-centered cube; (c)Kelvin; (d)Weaire-Phelan.

It is seen that the unit cell reconstruction method is simple and effective and can capture the structural characteristics of the porous media to a certain extent. However, this method ignores the randomness of the porous media such as the distribution of pore size. Moreover, some unit cells, such as simple cube and Kelvin structure, construct isotropic porous structures which are different from real anisotropic porous foams.

3.1.1.2. Random reconstruction method. In order to introduce randomness in the reconstruction process of the porous media, several random reconstruction methods have been proposed, and the representative one is Voronoi tessellation, which is a method of dividing a plane or space into specific areas based on a set of specific points. The distance from any position in each area to the specific point is smaller than the distance to the specific point in other areas. The areas in the Voronoi tessellation method could be viewed as cells in the porous media, and the random distribution creates the randomness of the reconstructed porous model. Based on the random point generation method and whether the random points are weighted or not, different Voronoi tessellations such as Laguerre-Voronoi tessellation and Poisson-Voronoi tessellation were built. The Laguerre-Voronoi tessellation method assigns weights for each points which can be used to control the distribution of pore diameter [62], and the reconstruction procedure based on this method is illustrated in Fig. 8. Firstly, the lognormal distribution was used to set the pore size distribution of the porous media. Subsequently, spheres with varying diameters were randomly and closely generated in the space based on the pore size distribution. Finally, the Laguerre-Voronoi tessellation of these random spheres was obtained, the solid structure and the surface structure were deleted, and the remaining boundary structures formed the porous skeleton. The study conducted by Wejrzanowski et al. [63] pointed out that the average number of faces per foam cell of the reconstructed porous models was in the range of 13.7-14.5, which was close to that of the actual porous media. The study of Nie

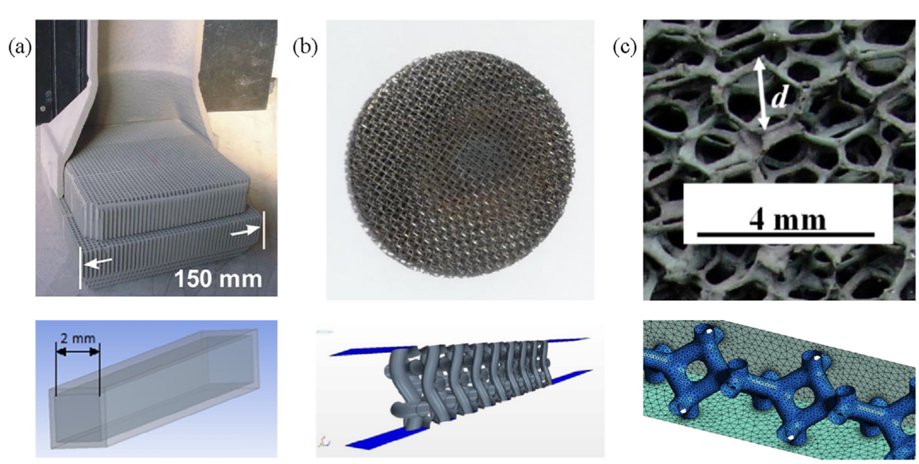

Fig. 6. Pore-scale models and porous volumetric solar receivers (a)Honeycomb (Adapted with permission [48]. Copyright 2013, Elsevier [49]; Copyright 2017, Elsevier); (b)Wire mesh (Reproduced with permission [50]. Copyright 2014, Elsevier [51]; Copyright 2018, Elsevier); (c)Foam (Adapted with permission [52]. Copyright 2011, Elsevier).

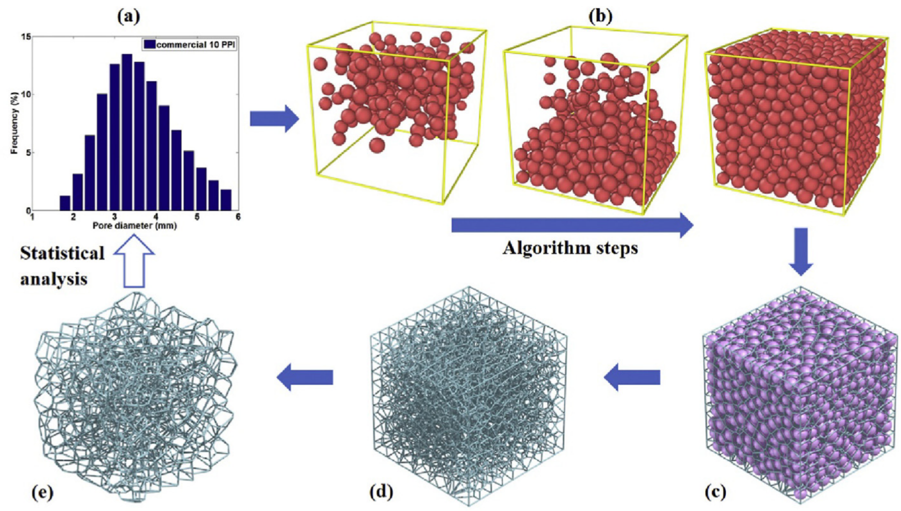

Fig. 8. Procedures for foam structure generation based on the Laguerre-Voronoi tessellation (Reproduced with permission [62]. Copyright 2017, Elsevier).

et al. [62] showed that different lognormal distributions could effectively control the cell volume of the reconstructed porous media. The average porosity and specific surface area can be adjusted by changing the ratio of skeleton thickness and sphere diameter.

Furthermore, different random construction methods were proposed based on the structural characteristics of the porous media. Kırca et al. [64] proposed a reconstruction method using the random sphere generation for carbon foam porous media. Similar to the Laguerre-Voronoi tessellation reconstruction process, a closely packed random spheres were firstly generated, and structural parameters such as the porosity and pore size distribution of the porous media were adjusted by controlling the sphere center distance and the sphere diameter distribution. Subsequently, the closely packed spheres were directly subtracted from the cube space, and the remaining part could be regarded as the porous skeleton, as illustrated in Fig. 9(a).

Considering for the conciseness of the Kelvin structure and its lack of randomness, Habisreuther et al. [65] proposed a random Kelvin structure by relocating the nodes position of the ordered structure. Fig. 9(b) compared the regular and random Kelvin structures. The node position was stochastically relocated but the maximum distance was limited to retain the original topological structure. However, compared with the porous model reconstructed by Magnetic Resonance Imaging (MRI), Habisreuther et al. found that the calculated pressure drop and tortuosity in the random Kelvin structure was relatively small. Besides, numerical methods such as multiple-point statistics [66], quartet structure generation set [67] and other random reconstruction methods [68] have been proposed for the reconstruction of the porous media with low porosity such as rocks. These methods are not used in the field of porous volumetric solar receivers and detailed description is no longer carried out.

3.1.1.3. X-ray CT reconstruction method. With the application of imaging technology and the improvement of computing capability, it is possible to accurately capture and digitize the real three-dimensional structure of the porous media. The X-ray computed tomography (CT) technology is widely applied which is a nondestructive imaging technique with high spatial resolution. The porous media reconstruction method based on the X-ray CT scan is presented in Fig. 10. Firstly, the suitable CT scanner and scan resolution should be determined. For the porous volumetric solar receiver, the pore diameter and strut thickness are usually at the millimeter scale, and the reported scan resolution for the porous foam reconstruction were normally in the range of 15-40 μm [69–79]. Several studies tried to capture the finer structure of the porous skeleton and utilized a higher scan resolution. For example, the scan resolutions in the study of Ambrosio et al. [80] and Haussener et al. [81] were 4 μm and 0.37 μm, respectively. It should be pointed out that the high-resolution scan is not compatible with a larger scan area and efficient numerical calculations. Therefore, it is necessary to evaluate the size of the porous media required for the numerical calculation and the current calculation capability, and carefully choose the scan resolution. The reconstructed

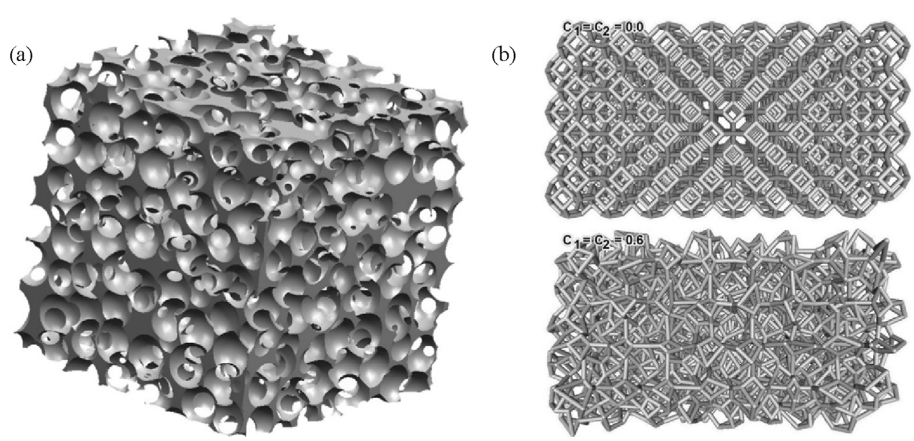

Fig. 9. Random reconstruction methods (a)Random spherical bubbles method (Reproduced with permission [64]. Copyright 2007, Elsevier); (b)Random Kelvin (Reproduced with permission [65]. Copyright 2009, Elsevier).

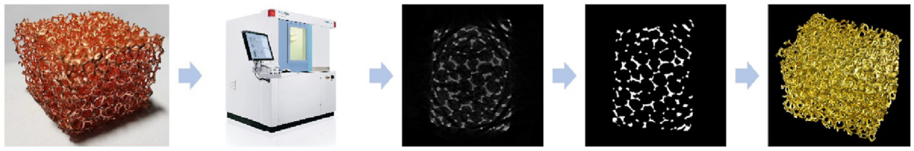

Fig. 10. Procedure of porous media reconstruction based on the X-ray computed tomography technique.

porous media for numerical calculation in literature have been tens of millimeters. Subsequently, the image processing such as noise reduction, sharpening, and contrast enhancement were needed for the scan images so as to better distinguish the porous skeleton from the background. These images could then be binarized and are ready for three-dimensional rendering. During the fabrication process of the porous media, the small hollow structures can be noticed at the center of the porous skeleton. In order to facilitate the subsequent mesh generation and numerical calculation, these small hollow structures need to be filled. Wang et al. [78] found that such filling processing had little effect on the structural parameters of the reconstructed porous media. The relative change of the porosity of the reconstructed porous media before and after filling was only 0.47%. Furthermore, the softwares such as Simpleware、VGStudio MAX、AVIZO、Mimics,and ImageJ could be applied for three-dimensional rendering and geometry repair. Finally, the three-dimensional model of the porous media could be exported to mesh generation softwares such as ANSYS ICEM and Simpleware ScanFE.

Based on the above studies, periodic unit cell reconstruction method can be directly selected for the reconstruction of honeycomb and wire mesh structures. For the porous foam, the periodic unit cell reconstruction method and random generation method can capture some of the structural characteristics of the porous structures and complexity of the reconstruction process is relatively low. The X-ray CT reconstruction method restores a more realistic three-dimensional structure of the porous media, but the reconstruction steps are relatively cumbersome and require higher computational resources.

3.1.2. *Direct pore-scale numerical simulation of fluid flow and heat transfer*

Based on the reconstructed porous models, the fluid flow and heat transfer inside the porous media could be directly solved at pore-scale. Direct pore-scale numerical simulation was mostly used in honeycomb porous volumetric solar receivers, and a few studies have investigated the porous volumetric solar receiver with packed bed and foam structures. The solar radiation transport and fluid flow and heat transfer processes inside the receivers are usually solved respectively because air is considered as a non-participating material and the optical properties of the porous media in the solar and infrared spectra are different. In the studies of Nakakura et al. [82,83], the radiative properties such as emissivity of the porous media were considered as constants and irrelevant to the spectrum. The incident solar radiation was adopted as the inlet boundary condition to the radiative transfer equation. Taking into account the solution methods of most studies, the solar radiation transport and fluid flow and heat transfer processes are introduced respectively.

3.1.2.1. Solar radiation transport in porous volumetric solar receivers. Porous volumetric solar receivers are typically integrated into solar tower and solar dish CSP systems. The different layouts of the CSP systems such as the heliostat field distribution and focusing strategy lead to different solar radiation intensities and distributions on the aperture of the receivers. Taking the solar tower CSP system as an example, researchers have developed a series of methods such as Monte Carlo ray tracing (MCRT) method and convolution method to investigate the solar radiation transport in the heliostat field and inside the receiver [84]. Based on the geometrical optics assumption, the MCRT method treats the solar radiation transport as the behaviors of a series of random processes of a large number of independent photons. The solar radiation distribution could be obtained by counting the reflection, absorption and distribution characteristics of photons. The implementation of the MCRT method is flexible and suitable for complex geometries, and it is widely used in the field of CSP. A series of commercial and open source programs have been developed such as MIRVAL, SolTrace, Tonatiuh, and SPTOPTIC [84,85]. The research group of He has applied the MCRT method to develop the optical models of different CSP systems, and the incident solar flux distribution and the optical properties have been determined [86–89]. Using the MATLAB toolbox developed by CIEMAT, Sanchez et al. [90] tracing the solar radiation transport in the solar dish system with different rim angles. Cagnoli et al. [49] and Ali et al. [91] applied Tonatiuh to study the solar radiation transport in the solar furnace and the solar tower system, respectively.

Distinguished from tubular solar receivers, the incident solar radiation penetrates, transfers, reflects, and absorbs inside the porous volumetric solar receiver. These processes can also be studied by the MCRT method and the implementation steps are shown in Fig. 11.

Considering that the pore-scale model of the porous volumetric solar receiver is small, the studies adopted uniformly distribution solar radiation with perpendicular irradiation or incident at a certain angle [79,82,83,92]. Subsequently, a photon goes through several processes such as reflection and absorption until it is no longer exist. The reflection describes the interaction of a photon with the porous skeleton and both constant reflectivity and local reflectivity models can be used to update the position and direction of the photon. The local reflectivity model applies Snell's law and Fresnel equations to calculate the reflectivity, and determines whether the photon is absorbed or not by comparing the local reflectivity to a random number [93,94]. The updated direction of the photon can be calculated based on diffuse or specular reflection laws. As for the honeycomb porous volumetric solar receiver, Sanchez et al. [90] and Cagnoli et al. [49] compared solar radiation distribution and optical properties with diffuse and specular channels. The diffuse walls aggravated the reflection loss in the front of the receiver, while the specular walls increased the penetration depth of solar radiation. These studies also pointed out that the actual reflection in the channels was a mixture of diffuse and specular reflections. As for the porous foam, Guévelou et al. [95] stated that the ratio of the surface roughness of the porous skeleton to the radiation wavelength determines the reflection characteristics of the radiation. When this ratio is low, the reflection can be regarded as specular. Du et al. [79] applied the MCRT method to determine the solar flux distribution insider the porous volumetric solar receiver. The inlet distribution of solar radiation was simplified to be uniform. Furthermore, Navalho and Pereira [15] investigated the solar radiation transport from the solar dish concentrator to the porous volumetric solar receiver, and the solar flux distribution were presented in Fig. 12. In order to simplify the ray tracing process, some studies treated the porous media as a uniform participating material and applied Beer's law to determine the solar radiation distribution [48,96].

3.1.2.2. Fluid flow and heat transfer in porous volumetric solar receivers. The direct pore-scale numerical simulation method does not introduce any empirical parameters and the detailed fluid flow and heat transfer processes could be captured by solving the governing equations.

For the fluid flow process, the surface of the porous skeleton forms the fluid channel which is usually set as non-slip boundary in the numerical model. The inlet boundary condition of the receiver is usually set with constant and uniform velocity or mass flow rate. For the flow state inside

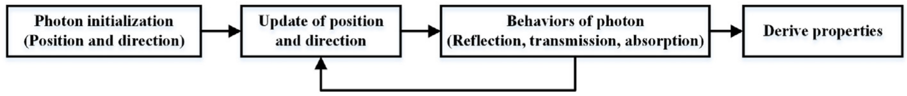

Fig. 11. Procedure of Monte Carlo ray tracing.

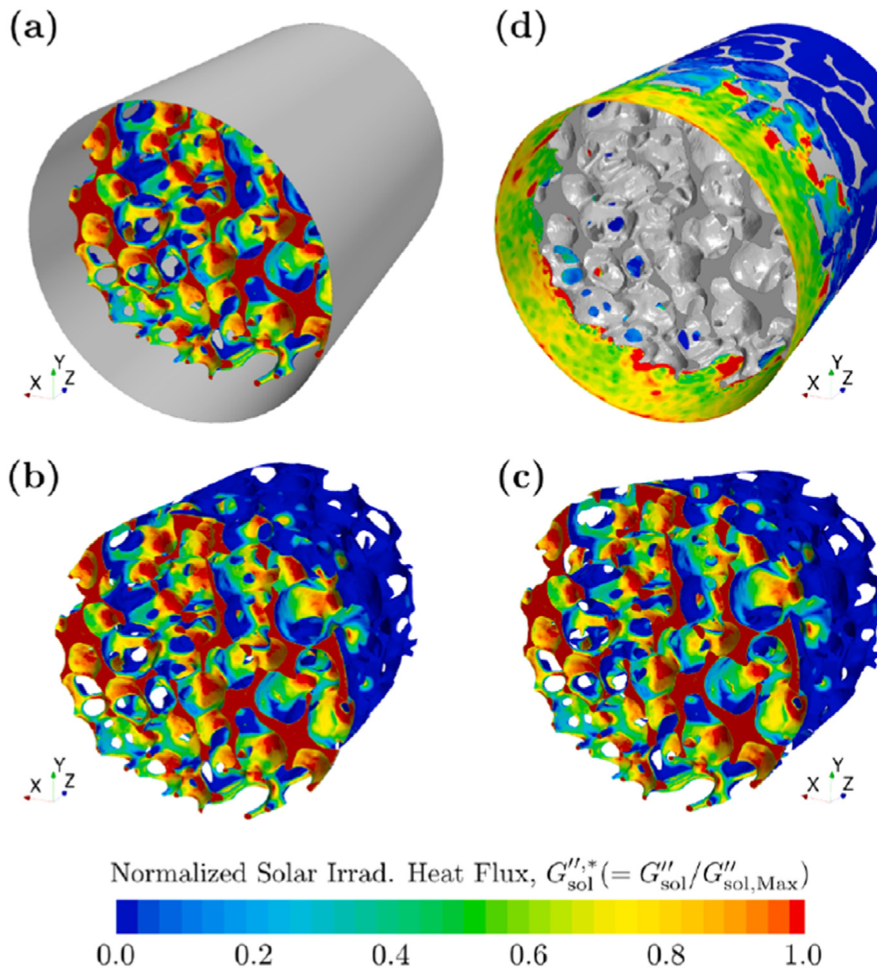

Fig. 12. Normalized solar irradiation heat flux distribution on the receiver (Reproduced with permission [15]. Copyright 2020, Elsevier).

the receiver, the turbulent model could be added according to the simulation conditions. For example, the honeycomb porous volumetric solar receiver has straight flow channels and the critical Reynolds number from laminar flow to turbulent flow is 2300 [49]. Moreover, Dybbs and Edwards [97] conducted visualization studies for the packed spheres and rods porous structures, and divided four flow regimes as follows: Darcy flow ($Re_d < 1$), inertia flow (1-10$<Re_d < 150$), unsteady laminar flow (150$<Re_d < 300$) and fully turbulent flow ($Re_d > 300$).

For the heat transfer process, the porous skeleton is irradiated by solar radiation and exchanges heat to the heat transfer fluid. Due to the non-uniform distribution and elevated temperature, the thermal radiation transfer in the porous skeleton should also be considered. The inlet air temperature is set to be constant and the solar flux calculated by the MCRT method is treated as heat flux on the surface or energy source in the porous skeleton. For example, Sanchez et al. [90], Cagnoli et al. [49] and Zhu and Xuan [92] treated the solar flux as second-type boundary condition on the surface of the porous skeleton. Ali et al. [91] extracted the volume mesh and set solar radiation as the energy source to the energy governing equation of the porous media. For radiative heat transfer, the heat transfer fluid (air) is a non-participating material and the radiative loss and radiative heat exchange between porous skeletons should be studied. The commonly used radiation models in porous volumetric solar receivers are Surface-to-Surface (S2S) model [49,91] and Discrete Ordinates (DO) method [82,83].

Based on the above direct pore-scale numerical method, Cagnoli et al. [49] analyzed the influence of tilt angle, channel size, and inlet air velocity on the optical and thermal performances of the honeycomb porous volumetric solar receiver. The typical temperature field of the honeycomb porous volumetric solar receiver was shown in Fig. 13(a). The variation of tilt angle of incident solar radiation led to different solar flux distributions in the channels but it has little effect on the thermal efficiency. Higher inlet air velocity and larger channel size were beneficial for improving the thermal efficiency. Zhu and Xuan [92] investigated the packed bed porous volumetric solar receiver with different unit cells. The MCRT method was applied to analyze the solar radiation penetration and the influence of incident angle. Due to lower porosity and large absorptivity of the porous material, the penetration depth was limited. The body centered cubic packed bed had smaller porosity which enhanced the convective heat transfer but increased the pressure resistance. The thermal efficiency of this type of porous volumetric solar receivers was relatively higher due to the lower temperature in the front surface of the receiver. The temperature fields under different inlet velocities were

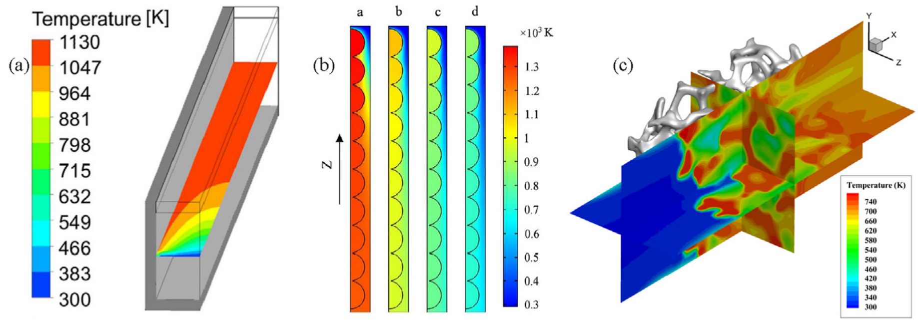

Fig. 13. Temperature fields in porous volumetric solar receivers (a)Honeycomb (Reproduced with permission [49]. Copyright 2017, Elsevier); (b)Packed bed (Reproduced with permission [92]. Copyright 2017, Elsevier); (c)Foam (Reproduced with permission [79]. Copyright 2017, Elsevier).

presented in Fig. 13(b). Du et al. [79] utilized the X-ray CT scan reconstruction method and built the real three-dimensional model of the porous foam. The MCRT method was applied to calculate the solar flux distribution inside the porous media which was then treated as the source term to the energy conservation equation. By solving the governing equations of fluid low and heat transfer, the detailed temperature field in the porous volumetric solar receiver was obtained in Fig. 13(c). Furthermore, based on the X-ray CT scan and image processing techniques, the porous models with controllable geometrical parameters were reconstructed by Du et al. [98]. The influence of geometrical parameters on the performance of the porous volumetric solar receiver was studied, and the Nusselt number correlation was derived.

3.2. Volume averaging simulation method

Different from the direct pore-scale numerical simulation method, the volume averaging simulation method does not consider the detailed structure of the porous media. It views the porous zone as two continuous solid phase and liquid phase regions, which represent the porous skeleton and the fluid, respectively. The interaction between the porous media and the fluid is introduced by effective properties and source terms in the governing equations. The volume averaging simulation method requires less computational resources and has been widely used to analyze and optimize the porous volumetric solar receiver. The accuracy of the volume averaging simulation method is greatly influenced by the selection of empirical parameters, and the determination of these parameters and correlations are presented as follows.

3.2.1. Fluid flow in the porous media

The porous skeleton impedes the fluid flow and flow resistance should be determined in the porous media. Darcy's law [99] is the theoretical basis which links the pressure drop with the superficial velocity by introducing permeability, as shown in Equation (1).

$$\frac{\Delta p}{\Delta L} = -\frac{\mu}{K}U \tag{1}$$

where p is the pressure, μ is the dynamic viscosity, K represents the permeability of the porous media, and U is the superficial velocity. Permeability reflects the capacity of porous media that allows the fluid to pass through, and the larger value means that the fluid could more easily flow in the porous media. Darcy's law is suitable for low Reynolds number fluid flow, and the inertia force becomes evident with the increase of flow velocity. Many equations have been proposed by considering the inertia effect based on Darcy's law, such as Darcy-Forchheimer equation as shown in Equation (2) and Darcy-Brinkman equation [99].

$$\frac{\Delta p}{\Delta L} = -\frac{\mu}{K}U - \frac{\rho c_{\mathrm{f}}}{\sqrt{K}}U^2 \tag{2}$$

Ergun and Orning [100] considered the Poiseuille equation and capillary flow and theoretically derived the pressure drop correlation of the packed bed porous media. Based on the experimental data of various fluids in different structures of the porous media, Ergun [101] determined the empirical parameters of the pressure drop equation, as shown in Equation (3).

$$\frac{\Delta p}{\Delta L} = 150\frac{(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu U}{D_{\mathrm{P}}^2} + 1.75\frac{(1-\varepsilon)}{\varepsilon^3}\frac{\rho U^2}{D_{\mathrm{P}}} \tag{3}$$

In the equation, ε and D_{P} are porosity and effective particle diameter of the porous media, respectively. This equation was derived for the porous media with small porosity but many researchers still adopted it to predict the pressure drop for the porous foam.

In order to characterize the pressure drop and calculate the permeability, experimental measurement and direct pore-scale numerical simulation method could be applied. Taking the research from Dietrich [102] as an example, a typical experimental set-up was shown in Fig. 14, which mainly includes a test sample, flow control devices, and data acquisition devices. During the test, the flow velocity was controlled by the blower and the pressure drop was measured by a sensitive manometer. By adding a heating device, the fluid temperature in the set-up could be precisely controlled, and the thermophysical properties could be calculated. Based on the derived data, the relationship between pressure drop and velocity could be established and the empirical parameters in the correlation could be derived. Madani et al. [103] pointed out that the inlet and outlet effects of the porous media influenced the pressure measurement. It was suggested to arrange the pressure measuring device inside the porous media along the fluid direction for precise measurements.

In Section 3.1, it is noticed that the direct pore-scale numerical method can obtain the detailed velocity field inside the porous media. Therefore, this method can be used to capture the pressure drop and velocity. Diani et al. [74] applied the X-ray CT reconstruction method and built porous foam models with different pore densities. The fluid flow of air inside the porous foams was simulated and the typical flow field was shown in Fig. 15(a). Wu et al. [104] used the periodic unit cell reconstruction method and built Kelvin models with different geometrical parameters. The pressure drop of different porous models was summarized in Fig. 15(b), and the relationship between pressure, velocity, porosity, and thermophysical properties of air was established. In the velocity range of 0.5-5 m/s, the correlation agreed well with the experimental data. Similarly, Nie et al. [105] reconstructed the porous models with the Laguerre-Voronoi tessellation, a pressure drop correlation with a larger applicable range of pore density and porosity was built.

Although many studies have been conducted for the pressure drop of the porous media, but different correlations deviate with each other and a direct comparison is hard. The main reasons for such a problem are summarized as follows: the measurement and data processing methods, and the selection of characteristic lengths of the porous media are not consistent, and the geometrical parameters of the porous media are not in the same range. Firstly, the pressure is usually measured at the inlet and outlet of the porous media. Considering that the thickness along the flow

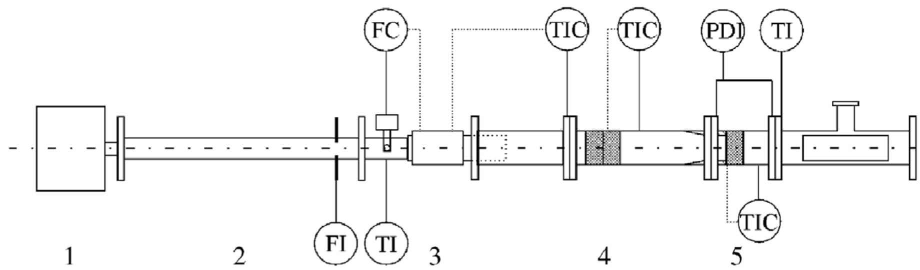

Fig. 14. Schematic of the experimental set-up for pressure drop measurements (Reproduced with permission [102]. Copyright 2009, Elsevier).

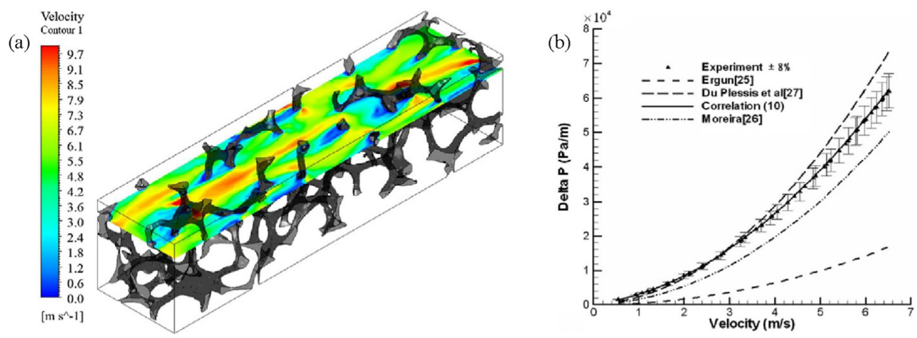

Fig. 15. Fluid flow and pressure drop characteristics based on the direct pore-scale numerical simulation (a)Flow field (Reproduced with permission [74]. Copyright 2015, Elsevier); (b)Pressure drop characteristic (Reproduced with permission [104]. Copyright 2010, Elsevier).

direction of the porous media is different in these studies, it is difficult to evaluate the difference in measurements caused by the inlet and outlet effects. Secondly, Kumar and Topin [106] summarized many studies and pointed out that correlating the experimental data by Darcy-Forchheimer equation without distinguishing the flow regime causes errors in the calculation of empirical parameters. The appropriate way to fit the experimental data is to find the seepage flow regime at first. Finally, the characteristic lengths of the porous media were selected differently in the studies. The commonly selected characteristic lengths include pore diameter, hydraulic diameter, and square root of permeability, which makes the direct comparison between different correlations more difficult. It should be noted that the problem of characteristic length selection also exists for the correlations of convective heat transfer coefficients. The proposed pressure drop correlations were thoroughly summarized in Refs. [106,107], and the correlations used for porous volumetric solar receivers are presented in Table 1.

From Tables 1 and it is seen that the Ergun equation was also used for the porous volumetric solar receiver despite that it was derived for the porous media with small porosity. In order to retain the form of the Ergun equation, Lacroix et al. [121] modified it for the porous foam by substituting the particle diameter in the original equation. The modified equation was suitable for the porosity in the range of 0.75-0.92, which agreed well with the experimental data when the velocity was under 6 m/s. Dietrich et al. [102] conducted experiments for the ceramic porous samples with different pore densities and porosities. The derived pressure drop correlation was compared with more than 2500 data in references and derivation was within ±40% [123]. The most widely used pressure drop correlation for the porous volumetric solar receiver was proposed by Wu et al. [104]. The Kelvin structures with different geometrical parameters were built, and the direct pore-scale numerical simulation method was applied. One good advantage for Wu's equation was the large applicable range of porosity and the Reynolds number.

Table 1
Pressure drop correlations of porous volumetric solar receivers.

Authors	Correlation	Application	Citing
Ergun [101]	$\frac{\Delta p}{\Delta L} = 150\frac{(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu U}{D_P^2} + 1.75\frac{(1-\varepsilon)}{\varepsilon^3}\frac{\rho U^2}{D_P}$	Derived for the packed porous media with low porosity	[108–111]
Wu et al. [104]	$\frac{\Delta p}{\Delta L} = \frac{(1039-1002\varepsilon)}{d_c^2}\mu U + \frac{0.5138\varepsilon^{-5.739}}{d_c}\rho U^2$	$10<Re<400$ $0.66<\varepsilon<0.93$	[112–120]
Lacroix et al. [121]	$\frac{\Delta p}{\Delta L} = 150\frac{(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu U}{d_P^2} + 1.75\frac{(1-\varepsilon)}{\varepsilon^3}\frac{\rho U^2}{d_P}$	Ergun equation with modified characteristic length	[122]
Dietrich et al. [102]	$\frac{\Delta p}{\Delta L} = 110\frac{\mu}{\varepsilon \cdot d_h^2}U + 1.45\frac{\rho}{\varepsilon^2 \cdot d_h}U^2$	$10<Re<3900$	[18]

3.2.2. Heat transfer in the porous media

In order to obtain the heat transfer characteristics in the porous media, both the local thermal equilibrium (LTE) model and the local thermal non-equilibrium (LTNE) model can be applied. The main difference between these two models is whether the temperature difference between the porous media and the heat transfer fluid exists or not. Considering for the large difference in thermal conductivity of air and porous media, the LTNE model has been widely accepted [124,125]. This model solves the energy conservations equations in both the porous media and the heat transfer fluid regions, and uses source terms to represent the energy exchange between the two regions.

3.2.2.1. Solar radiation transport in porous volumetric solar receivers. The process of solar radiation transport and the solar energy coupling method are summarized in Fig. 16. The solar radiation transport before entering the porous volumetric solar receiver can be calculated by the MCRT method, which has been introduced in the above section. For simplicity, some studies adopted uniform or Gaussian distribution of the incident solar radiation at the inlet of the porous volumetric solar receiver [112, 115,117,118,126,127].

When the solar radiation distribution in the aperture of the solar

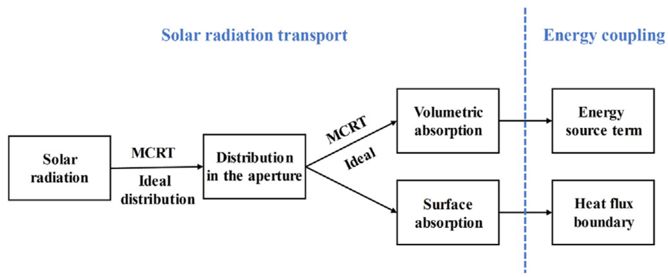

Fig. 16. Solar radiation transfer and solar energy coupling method.

receiver is determined, the solar radiation transport inside the porous media should be analyzed. Some studies considered that the optical thickness of the porous media is large, and the solar radiation absorption occurs within a short distance from the front surface of the receiver. Therefore, solar radiation is simplified as the heat flux boundary condition for the energy conservation equation or radiative transfer equation [108,110,111,113]. On the other hand, some researchers thought that it was necessary to consider the solar radiation penetration process. The MCRT method and the ideal exponential decay based on Beer's law could be applied, and the solar energy inside the porous volumetric solar receiver were treated as a source term to the energy conservation equation or radiative transfer equation [18,109,112,114,115,117,119,120,126].

Different from the implementation of MCRT in the direct pore-scale numerical simulation method, the MCRT method in the volume averaging simulation method does not consider the real interaction between the porous skeleton and photon. The photon propagation length and scattering models are implemented with random processes which correspond to the radiative properties of the porous media and the heat transfer fluid. In order to determine the propagation length Δs of the photon between two collisions, the extinction coefficient β and the random number ξ are used to establish the relationship as Δs=-lnξ/β. The extinction coefficient could be measured by a spectrometer or calculated by the direct pore-scale numerical simulation. One typical correlation for the extinction coefficient of the porous media is shown in Equation (4). Based on the geometric optics approximation [128] and the experimental data by Hendricks and Howell [129], the constant coefficient in Equation (4) is 3 and 4.8, respectively.

$$\beta = \frac{C(1-\varepsilon)}{d_{\mathrm{p}}} \tag{4}$$

The scattering process inside the porous media is determined by the scattering phase function, which gives the method to calculate the azimuthal angle. Taking the Henyey-Greenstein phase function as an example, the equation for the azimuthal angle with anisotropic scattering is given by Equation (5). The range for asymmetry factor g is $-1\sim1$, and value of 0 means the isotropic scattering and Equation (6) should be used [130]. Chen et al. [131] and Qiu et al. [132] treated the porous volumetric solar receiver as the uniform scattering material and investigated the solar radiation transport process and the influence of geometrical parameters on the solar flux distribution. Barreto et al. [133] solved the inverse problem with experimental data of hemispherical diffuse reflectance, and it was found that that the solar radiation scattering in the SiC ceramic foams is slightly backwards and the best value of asymmetry factor was -0.25. The detailed photon transport inside the porous media could refer to Refs. [134,135].

$$\cos\theta = \frac{1}{2g}\left\{1+g^2-\left[\frac{1-g^2}{1-g+2g\xi}\right]^2\right\} \tag{5}$$

$$\cos\theta = 2\xi - 1 \tag{6}$$

Chen et al. [116] compared the thermal performances of the porous volumetric solar receiver when treating the solar radiation as the heat flux boundary condition or energy source term. The volumetric absorption was ignored when using heat flux boundary condition which leads to a higher surface temperature and larger thermal radiative losses. It was also shown that the temperature distribution was slightly influenced by the incident angle of the solar radiation, and the largest deviation in temperature was 3.4%.

3.2.2.2. *Coupled heat transfer in porous volumetric solar receivers*

3.2.2.2.1. Heat conduction. According to the LTNE model, the effective thermal conductivities of the porous media and the heat transfer fluid are necessary for simulating the heat conduction. The effective thermal conductivities of packed beds and porous foams have been widely investigated and the commonly used methods include asymptotic models, experimental measurements, theoretical models and direct pore-scale numerical simulation.

Based on the one dimensional heat conduction assumption, the most commonly used parallel and series models of thermal conductivities are present in Equation (7) and Equation (8), which show the lower and upper bounds for the thermal conductivity of the porous media. In the parallel model, the contribution of the porous media and the heat transfer fluid can be easily separated and it has been adopted in many studies of porous volumetric solar receivers. Besides, many other asymptotic models were proposed such as the Maxwell-Eucken model and the Miller model [136]. The asymptotic models not only give the estimation of upper and lower bounds but also provide ideas for the construction of effective thermal conductivity correlations. For example, Calmidi and Mahajan [137] modified the Equation (7) and introduced two coefficients for better fitting with the experimental data. Dul'nev and Zarichnyak as well as Chaudhary and Bhandari combined the parallel and series models linearly and by production [136].

$$\lambda_{\parallel} = \varepsilon\lambda_{\mathrm{f}} + (1-\varepsilon)\lambda_{\mathrm{s}} \tag{7}$$

$$\lambda_{\perp} = \left(\frac{\varepsilon}{\lambda_{\mathrm{f}}} + \frac{1-\varepsilon}{\lambda_{\mathrm{s}}}\right)^{-1} \tag{8}$$

Secondly, experimental measurements can give an accurate value of the effective thermal conductivity based on Fourier's law. Fig. 17 presents the typical experimental set-up for the effective thermal conductivity measurement in the study of Paek et al. [138]. The experimental set-ups used the water bath to generate the temperature difference on the two sides of the porous media, and uses thermocouples to measure the temperatures at different positions in the test sections. In order to acquire the heat flux passes through the porous media, Calmidi and Mahajan [137] assumed that the heat loss could be neglect due to the thermal insulation material and all the heat generated by the electric heating flowed through the porous media. Bhattacharya et al. [139] further measured

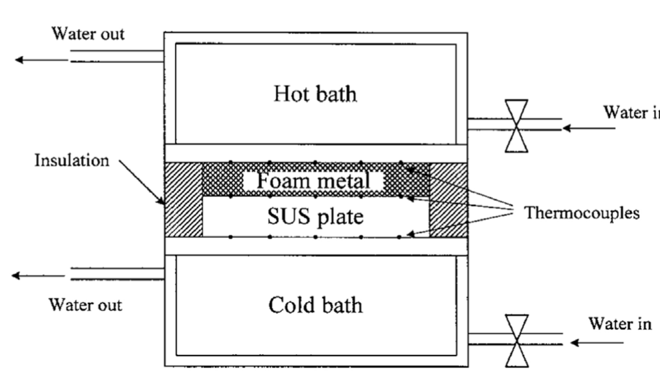

Fig. 17. Schematic of experimental set-ups for effective thermal conductivity measurements of Paek et al. (Reproduced with permission [138]. Copyright 2000, Springer Nature).

the temperature distribution inside the thermal insulation material and estimated the heat loss to the environment. Paek et al. [138] and Schmierer et al. [140] arranged a stainless steel plate with known thermal conductivity and the porous sample in series. Based on the assumption of heat flux continuity, the temperatures on the two sides of the stainless steel plate could be used to calculate the heat flux flowing through the porous media. It should be mentioned that the thermal contact resistance influences the experimental measurement due to the high porosity and irregular structure of the porous media. Sadeghi et al. [141] measured the effective thermal conductivity of porous samples with different thicknesses, and the thermal contact resistance could be determined by assuming that it is constant. Fiedler et al. [142] further pointed out that the thermal contact resistance could change under different experimental conditions. It could be accurately calculated by analyzing the experimental data with numerical simulation. Moreover, Vicente et al. [143] derived the effective thermal conductivity by capturing the temperature response of the porous media to the radiative heat flux by a thermal imaging camera. Some studies directly measured the effective thermal conductivity by the transient plane source technique with a hot disk [144–146].

Thirdly, the theoretical models simplify the porous media to the arrays of periodic unit cells, such as hexagons in the two-dimensional space and the Kelvin structure in the three-dimensional space. The idealized porous structures could be divided into multiple zones along the heat flux direction and the theory of thermal resistance could be applied to calculate the effective thermal conductivity. Calmidi and Mahajan [137] simplified the porous media with hexagonal periodic units and a square block structure was added at the vertices of the hexagon to reflect the larger thickness in the connection of the porous skeleton. The representative unit in the hexagonal structure was extracted and the porous structure was divided into multiple layers along the heat flux direction. In each layer, the porous skeleton was in parallel with the heat transfer fluid, and the effective thermal conductivity was calculated by Equation (7). The effective thermal conductivity of the entire porous media was the serial combination of multiple layers which was calculated with Equation (8). Finally, the calculated effective thermal conductivity was compared with experimental data to determine the value of undetermined constants in the correlation. Based on this study, some researchers further modified the shape of the block in the skeleton intersection [136] which was not presented in detail.

Lastly, the direct pore-scale numerical simulation is an effective method to calculate the effective thermal conductivity of the porous media because the temperature field inside the porous media can be easily obtained. Many studies investigated the effective thermal conductivity of the porous media with different geometries [57,105,147].

Despite that many correlations for the effective thermal conductivity have been established, there still exist certain limitations. Kumar and Topin [148] as well as Dietrich et al. [149]pointed out that the thermal conductivity of the porous skeleton was different from the intrinsic thermal conductivity of the material that make up the porous media. Due to different processing techniques and impurities, deviations exist between the thermophysical properties of the framework and the intrinsic thermophysical properties of the material. Moreover, Ranut [136] found that using a detailed and complex structure of the porous medium model does not guarantee a more accurate prediction of the equivalent thermal conductivity. Introducing the undetermined constants in the correlation was beneficial for the curve fitting with the experimental data. Finally, some correlations have a mutual term of both the heat transfer fluid and porous media. When it is hard to distinguish the contribution of the porous media and the heat transfer fluid to the thermal conductivity, such correlations can not be applied for the LTNE model.

Based on the above analysis, the studies for porous volumetric solar receivers generally used the parallel model or the Schuetz-Glicksman correlation in the LTNE model [150], which are presented in Table 2. It should be noted that the original Schuetz-Glicksman correlation was proposed for the porous media with large porosity and the contribution of the heat transfer fluid to the thermal conductivity was not considered [151].

Table 2
Effective thermal conductivity correlations of porous volumetric solar receivers.

Models	$\lambda_{f,eff}$	$\lambda_{s,eff}$	Citing
Parallel model	$\varepsilon \cdot \lambda_f$	$(1-\varepsilon) \cdot \lambda_s$	[108,110,114]
Schuetz-Glicksman correlation [150]	$\varepsilon \cdot \lambda_f$	$(1-\varepsilon) \cdot \lambda_s / 3$	[18,111–113, 115–120]

3.2.2.2.2. Convective heat transfer. In the volume averaging simulation method, the convective heat transfer between the porous media and the heat transfer fluid is set as a source term in the energy conservation equations. Therefore, accurate evaluation of the convective heat transfer coefficient has become the key to investigate the convective heat transfer process in the porous media. The commonly used methods include the single blow method and the direct pore-scale numerical simulation method.

The basic principle of the single blow method is to measure the temperature variation of the fluid when it flows through the porous media with different temperatures, and the inverse problem is solved based on experimental data and numerical simulation. In the experiments, the inlet and outlet fluid temperatures are monitored and the experimental process can be either heating or cooling the fluid. An experimental set-up for the porous media cooling the high temperature fluid is presented in Fig. 18(a). In the first stage of the experiment, the heat transfer fluid passed through the heating device and the test section. The experimental system reached a steady state and the temperature of the porous media and the heat transfer fluid was no longer changed. In the second stage, the high temperature porous sample was quickly replaced by un unheated porous sample. Due to the temperature difference, the cooling process of the heat transfer fluid could be monitored by thermocouples at the outlet. Based on the experimental conditions, the transient LTNE model of the porous media with the heat transfer fluid was established, and the boundary conditions as well as the thermophysical properties maintained the same with the experiment. By adjusting the value of convective heat transfer coefficient in the numerical model, the difference between the numerical simulation results and the experimental data was gradually reduced and finally the convective heat transfer coefficient was obtained.

The implementations of the single blow method are different in the studies. In order to adjust the temperature of the heat transfer fluid, many researchers adopted electric heating devices. But Giani et al. [154] mixed hydrogen into the fluid, and the hydrogen reacted with the catalyst to heat the fluid. Dietrich [153] designed the high temperature and low temperature flow paths, which could be switched to change the temperature of the fluid that passed through the porous media, as shown in Fig. 18(b). A similar two flow paths design was also used by Lu et al. [155]. Besides, many studies monitored both inlet and outlet temperatures of the heat transfer fluid and did not maintain the constant temperature inlet boundary condition [156–158]. In the numerical models, the measured fluid temperature in the inlet was set as the time-dependent boundary condition, and the calculated outlet temperature was used for the comparison with experimental data. The single blow method is a transient measurement method, and some researchers also applied steady methods to evaluate the convective heat transfer coefficient. Ichimiya [159] utilized the side wall to heat the porous sample and let it heat the low temperature heat transfer fluid. The temperature of the side wall and the fluid could be measured which were then used to calculate the convective heat transfer coefficient at the wall. This parameter was selected as the indicator in the inverse problem.

On the other hand, the direct pore-scale numerical simulation method has been widely used to investigate the convective heat transfer in the porous media. Iasiello et al. [71] compared the convective heat transfer coefficient of real porous foams with the Kelvin structure. When the two porous models had the similar porosity, cell diameter, and strut

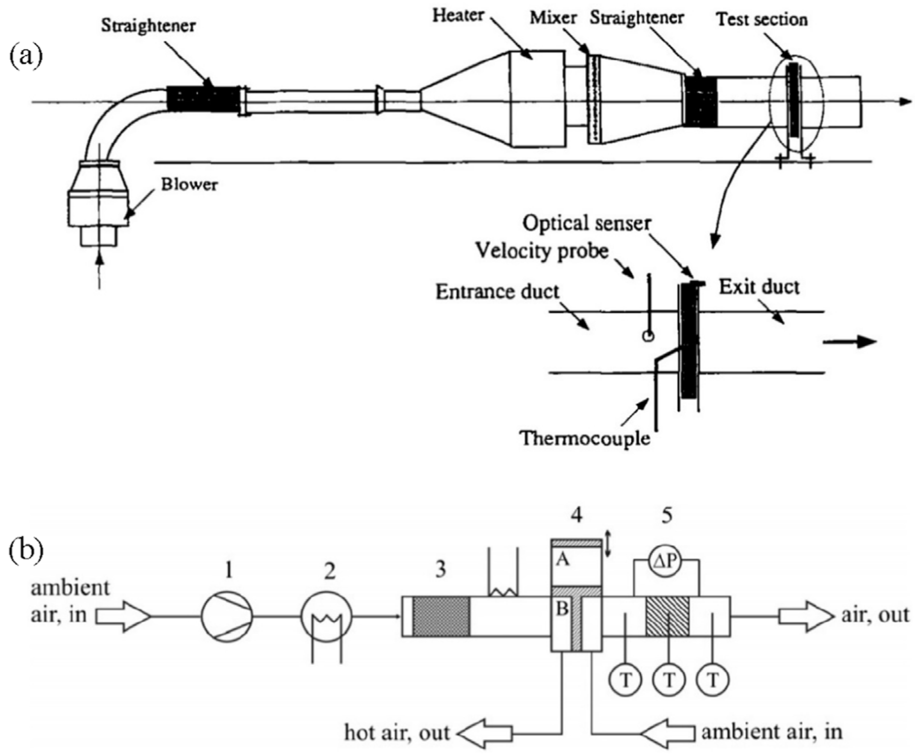

Fig. 18. Schematics of experimental setups of convective heat transfer coefficient measurements (a) Younis and Viskanta (Reproduced with permission [152]. Copyright 1993, Elsevier); (b)Dietrich (Reproduced with permission [153]. Copyright 2013, Elsevier).

thickness, the Kelvin structure had a larger volumetric convective heat transfer coefficient due to the larger specific surface area. Meinicke et al. [160] further compared the porous foam, the Kelvin structure, the Weaire-Phelan structure, the simple cube periodic structure. The porous foam had more tortuous flow channels which led to a larger convective heat transfer coefficient. It should be pointed out that the boundary conditions used in the direct pore-scale numerical simulations were different. Some studies [69,75,161] used the constant temperature boundary condition on the surface of the porous media, while other studies [71,74] adopted the constant heat flux boundary condition. Ranut et al. [72] compared the simulation results under these two boundary conditions and found that the constant heat flux boundary led to larger convective heat transfer coefficient under the same Reynolds number. Du et al. [162] further compared these two boundary conditions with the non-uniform solar flux distribution. It was shown that the calculated Nusselt number was a good approximation to that of the porous volumetric solar receiver when utilizing the constant temperature boundary condition.

Based on the above research methods, a large number of correlations of the convective heat transfer coefficients of the porous media have been proposed. However, differences in the correlation forms and applicable ranges still exist. Firstly, most studies adopted the correlation form as $Nu=f(Re,Pr)$, and some studies introduced the influence of porosity [52,152,156] and the ratio of thickness of the porous media and the pore size [152,163]. Secondly, as stated above, the selection of the characteristic length was not consistent which makes it difficult for direct comparison. Lastly, the LTNE model should be solved when applying single-blow method. The selection of other empirical parameters such as permeability and effective thermal conductivity influence the calculation of the convective heat transfer coefficient. Table 3 summarized the convective heat transfer coefficient correlations of porous volumetric solar receivers.

Wu et al. [52] and Du et al. [98] applied the direct pore-scale numerical simulation method and obtained the convective heat transfer coefficient for the porous foam. Dietrich et al. [153] conducted a single blow experiment for the porous foams and correlated the convective heat transfer coefficient with hydraulic diameter. The correlations of Hwang

Table 3
Convective heat transfer coefficient correlations of porous volumetric solar receivers.

Author	Correlation	Notes	Citing
Wu et al. [52]	$h=\lambda_f(32.504\varepsilon^{0.38}-109.94\varepsilon^{1.38}+166.65\varepsilon^{2.38}-86.98\varepsilon^{3.38})Re_c^{0.438}/d_c^2$	Porous foam $70<Re_c<800$ $0.66<\varepsilon<0.93$	[112–120]
Du et al. [98]	$Nu=2.785+1.261\cdot\varepsilon^{0.8859}\cdot Re_d^{0.5237}$	Porous foam $3<Re_d<233$ $0.74<\varepsilon<0.89$	[164,165]
Hwang et al. [166]	$h=\begin{cases}0.004\left(\frac{d_{void}}{d_{par}}\right)^{0.35}\left(\frac{k_f}{d_{par}}\right)Pr^{0.33}Re_{par}^{1.35}, Re_{par}\le 75\\ 1.064\left(\frac{k_f}{d_{par}}\right)Pr^{0.33}Re_{par}^{0.59}, Re_{par}\ge 300\end{cases}$	Packed bed Interpolation of h when $75<Re_{par}<300$	[109,110]
Wakao et al. [167]	$h=\lambda_f(2+1.1Pr^{1/3}Re^{0.6})/d_{par}$	Packed bed $15<Re_{par}<8500$	[111]
Dietrich et al. [153]	$h=\lambda_f(0.57\cdot C_{Re}\cdot C_{geo}\cdot Re_h^{2/3}Pr^{1/3})/d_h$	Porous foam $60<Re_h<1000$ $0.66<\varepsilon<0.93$	[18]

et al. [166] and Wakao et al. [167] were initially for the packed bed porous media but not for the porous volumetric solar receiver.

3.2.2.2.3. Radiative heat transfer. The radiative heat transfer becomes evident in solar receivers as the temperature goes higher. Because it is difficult to completely solve the radiation transfer equation, the simplifications should be made based on the radiative properties of the porous media and the heat transfer fluid. The commonly used methods include Rosseland approximation and P_1 model [168].

The porous volumetric solar receiver is generally considered as the optically thick semi-transparent medium, and the transmission distance of radiation inside it is relatively short. Therefore, the radiative heat exchange usually depends on the state of the adjacent area, and the radiative heat transfer tend to be the same as the heat conduction. Based on this characteristic, the Rosseland approximation is applied, the radiation transfer equation can be simplified, and the radiative heat flux is calculated by Equation (9) [168].

$$q_{\mathrm{r}} = -\frac{16n^2\sigma T_{\mathrm{s}}^3}{3\beta}\cdot\frac{\mathrm{d}T_{\mathrm{s}}}{\mathrm{d}x} = -\lambda_{\mathrm{r}}\cdot\frac{\mathrm{d}T_{\mathrm{s}}}{\mathrm{d}x} \tag{9}$$

where n is the refractive index, σ is the Stefan-Boltzmann constant which equals 5.676×10^{-5} W/($m^2\cdot K^4$), and λ_r stands for the equivalent radiative conductivity. Based on the Rosseland approximation, the influence of radiative heat transfer can be added to the heat conduction term in the energy conservation equation.

Some studies have pointed out that the application of the Rosseland approximation causes relatively large errors at the boundary [113]. Therefore, the P_1 model is applied which is a simplification of P_N model. The P_1 model considers the scattering process and the radiation transfer equation is shown by Equation (10) [112].

$$-\nabla\cdot\left(\frac{1}{3(a+\sigma_s)}\nabla G\right) = a\left(4\sigma T_s^4 - G\right) + S \tag{10}$$

In this equation, a and σ_s are the absorption and scattering coefficients, respectively. The radiative heat flux added to the energy conservation equation can be calculated by Equation (11).

$$-\nabla\cdot q_r = -a\left(4\sigma T_s^4 - G\right) \tag{11}$$

Except the extinction coefficient, the scattering coefficient is necessary for the calculation. Based on the geometric optics, the absorption and scattering coefficients depend on the geometrical parameters of the porous media and the emissivity, which are shown in Equation (12) and Equation (13) [112].

$$a = 1.5\varepsilon_{\mathrm{e}}(1-\varepsilon)/d_{\mathrm{p}} \tag{12}$$

$$\sigma_s = 1.5(2-\varepsilon_{\mathrm{e}})(1-\varepsilon)/d_{\mathrm{p}} \tag{13}$$

It should be noted that the Equation (10) is only valid for the uniform scattering, and an anisotropic phase function coefficient can be further considered [169]. Moreover, other radiation models such as the DO model could also be used for porous volumetric solar receivers.

Mey et al. [113] compared the thermal performances of porous volumetric solar receivers calculated by the Rosseland approximation, the P_1 model, and the two-flux approximation. When applying the Rosseland approximation, the solar flux was added as the heat flux boundary condition in the energy conservation equation. While in the P_1 model, the solar flux was treated as the heat flux boundary condition to the radiation transfer equation. It was found that the Rosseland approximation led to a high temperature at the inlet of the receiver but the temperature inside the receiver and at the outlet were close to that calculated by the two-flux approximation. Wang et al. [114] compared the Rosseland approximation and the modified P_1 model for porous volumetric solar receivers. When treating solar energy as the energy source term in the energy conservation equation, the Rosseland approximation and the modified P_1 model gave similar simulation results and the largest deviation was 4.97%. Zaversky et al. [18] compared the Rosseland approximation and the DO model and found that the Rosseland approximation was better when treating solar energy as the source term in the energy conservation equation. Therefore, the treatment of solar energy and the selection of radiation models both influence the calculation of radiation heat transfer.

Based on the volume averaging simulation method, a large number of studies have carried out to investigate the influence of geometrical parameters and operating conditions on the thermal performance of porous volumetric solar receivers. Wang et al. [114], Chen et al. [116] and Barreto et al. [119] investigated the solar radiation transport in the heliostat field coupled with the fluid flow and heat transfer processes inside porous volumetric solar receivers. Wu and Wang [170] as well as Wang et al. [171] established transient heat transfer models for porous volumetric solar receivers and analyzed the response time of the receivers under the transient heat flux boundaries with different thermophysical properties. Du et al. [127] coupled the genetic algorithm and the volume averaging simulation method which served as an efficient tool to optimize the geometrical parameters and operating conditions of porous volumetric solar receivers.

4. Advances in volumetric absorption enhancement

Based on above experimental and numerical methods, studies have been carried out for the performance optimization of porous volumetric solar receivers. The volumetric absorption enhancement and thermal efficient improvement of receivers could be realized by geometrical parameter optimization and spectrally selective absorption.

4.1. Geometrical parameter optimization

The porous media prepared by conventional processing technologies usually have a relatively uniform pore structures, and therefore have relatively uniform optical and radiative properties as well as uniform fluid flow and heat transfer characteristics. However, the energy transfer in porous volumetric solar receivers is non-uniform. For example, the penetration depth of solar radiation is limited due to the larger extinction coefficient of the porous media. The absorption of solar radiation occurs at the front of the receiver, which is contrary to the concept of volumetric absorption by the porous media. Secondly, the incident solar radiation distribution is non-uniform due to the concentrating process in the heliostat field. The uniform heat transfer capability of the porous media does not match up with the non-uniform solar energy, which leads to local overheating and deterioration in thermal efficiency. Therefore, the concept of functionally graded material is used in the design of porous volumetric solar receiver.

Multi-cavity volumetric receiver [42] was an earlier attempt that adjusted the local geometrical parameters to improve the performance of porous volumetric solar receivers. This receiver was built by horizontal foils and side-bars, which formed cavities to absorb solar radiation. In order to reduce the front surface area, a staggered layout of the front was proposed as shown in Fig. 19(a). However, the experimental studies did not indicate the performance improvement by the staggered design as compared with the uniform structure. Fend et al.2 combined the porous materials with different pore densities which had a larger pore density in the front as presented in Fig. 19(b). The smaller pore size (80 PPI) of the porous media could efficiently absorb the solar radiation while enhancing the convective heat transfer. The larger pore size (20 PPI) in the rear was beneficial for reducing the flow resistance. The double-layer design improved the thermal efficiency by 10%, as compared to the uniform porous media with a 20 PPI. Since then, the porous volumetric solar receivers with double, triple or multilayer structure have been extensively studied. Roldán et al. [126] and Chen et al. [115] established the numerical model for the triple-layer and double-layer porous volumetric solar receivers, respectively. Both studies revealed that the receiver with decreasing porosity distribution could improve the overall thermal efficiency, which outperforms the traditional uniform porous

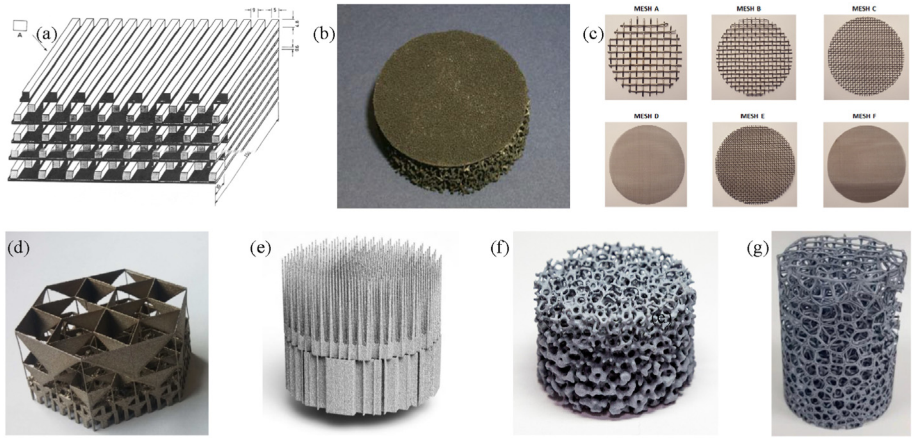

Fig. 19. Porous volumetric solar receivers with varying geometries along the axis (a)Multi-cavity (Reproduced with permission [42]. Copyright 1993, Elsevier); (b) Fend et al. (Adapted with permission [174]. Copyright 2004, Elsevier); (c) Avila-Marin et al. (Reproduced with permission [172]. Copyright 2018, Elsevier); (d)Luque et al. (Adapted with permission [175]. Copyright 2018, Elsevier); (e)Capuano et al. (Reproduced with permission [176]. Copyright 2017, Elsevier); (f)Du et al. (Reproduced with permission [165]. Copyright 2020, Springer Nature); (g)Avila-Marin et al. (Reproduced with permission [177]. Copyright 2022, Elsevier).

receivers with porosity of 0.7-0.9. The larger porosity in the front could increase the penetration depth of solar radiation, reduce the surface temperature, and limit the thermal radiative loss. Chen et al. further investigated the influence of thickness ratio of the two layers on the outlet temperature of the receiver. It is shown that the improvement in thermal efficiency disappeared when the thickness of the front layer reached a certain value. When the thickness of the front layer was too small, the volumetric absorption of t solar radiation was weakened. The optimized thickness ratio for the double-layer porous volumetric solar receiver was 3:7. Avila-Marin et al. [172,173] and Zaversky et al. [18] conducted experimental studies for the triple-layer wire mesh and silicon carbide foam porous volumetric solar receivers, as presented in Fig. 19(c). Avila-Marin et al. found that the optimized double-layer receiver had a large front porosity and a large specific surface area in the rear. Besides, the triple-layer receiver could not further improve the thermal performance as compared with the optimized double-layer receiver. However, in the experiment of Zaversky et al., no evident improvement was observed by the double-layer or triple-layer designs. It was concluded that a careful selection of the single-layer configuration was sufficient to optimize the performance of the receiver. Based on the developed numerical model, the optimized thickness ratio calculated by Zaversky et al. was also 3:7, which was in agreement with the result of Chen et al. [115].

Furthermore, the porous volumetric solar receivers with multilayers and gradually-varied structures have also been proposed. Luque et al. [175] designed the hierarchically-layered fractal-like porous volumetric solar receivers based on additive manufacturing techniques as shown in Fig. 19(d). The results showed that the volumetric effect could be obtained but the thermal efficiency was relatively low. Capuano et al. [176] also applied additive manufacturing techniques and fabricated the porous media having a pin zone with gradually changed structures, as shown in Fig. 19(e). Compared with HiTRec-II with uniform honeycomb structure, the thermal performance was improved while the volumetric effect was obtained. Du et al. [165] developed a replica method with the multilayer recoating technique, and the silicon carbide porous ceramic with linear-changed geometrical parameters was fabricated as presented in Fig. 19(f). The experimental study showed that the thermal efficiency could be improved by 1.46% with the gradually-varied design, compared with the best uniform porous structure with an average porosity of 0.52. Based on the 3D printing and replica method, the uniform and gradual cell size porous receivers with different periodic cellular structures were fabricated and tested by Avila-Marin et al. [177], as shown in Fig. 19(g). The experimental results presented that Voronoi porous receiver with gradually increasing cell size exhibited better thermal performance due to lower solid temperature. Moreover, 3D printing technique has also been adopted by different researchers for the fabrication of porous receivers with graded pore geometries and complex lattice structures [178,179].

On the other hand, some studies focused on the porous volumetric solar receivers with radial graded structures. The radial graded porous media can improve the thermal performance of the receivers by adjusting the air mass flow distribution under non-uniform solar radiation. Roldán et al. [126] designed the porous volumetric solar receivers with decreasing or increasing porosity distribution and analyzed the performance with the LTE model. Arranging the porous structure with large porosity at the center of the receiver was beneficial to increase the average outlet temperature, but the temperature gradient of the receiver also increased. Chen et al. [117] proposed a porous volumetric solar receiver with composite structures. A small zone in the inlet near-wall region was arranged with the porous media having smaller porosity and pore size. The increased flow resistance compelled the fluid to move into the center region with a higher solar flux, and the solid temperature gradient in the front could be reduced. Based on the X-ray CT reconstruction and 3D printing techniques, Du et al. [164] experimentally investigated the fluid flow and heat transfer performances of the radially graded porous volumetric solar receiver, as shown in Fig. 20. The results showed that the thermal efficiency was relatively improved by 4.1% while the flow resistance decreased relatively by 8.6%, compared with the optimized uniform porous volumetric solar receiver with 20 PPI. By utilizing the volume averaging simulation method, the flow and temperature fields inside the porous volumetric solar receivers were visualized as illustrated in Fig. 21. It was found that the local overheating existed in the uniform porous volumetric solar receivers due to the fact that the air viscosity increases as the temperature goes higher. The smaller porosity at the center of the radially graded porous volumetric solar receiver decreased the flow resistance at the high temperature region, which led to more heat transfer fluid to absorb the concentrated solar radiation.

4.2. Spectrally selective absorption

From the viewpoint of energy balance, the key to improve the receiver thermal efficiency is to increase the solar radiation absorption and reduce energy losses. The thermal radiative loss at the front of the receivers becomes the major energy loss under high temperature operation condition. Therefore, many researchers tried to adjust the radiative properties of the materials and structure in order to increase the absorptivity in the solar spectrum while reducing the emissivity in the infrared spectrum. The commonly investigated spectrally selective

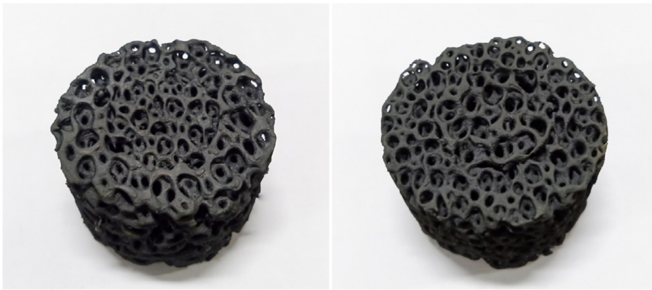

Fig. 20. Radially graded porous media fabricated by 3D printing (Adapted with permission [164]. Copyright 2020, Elsevier).

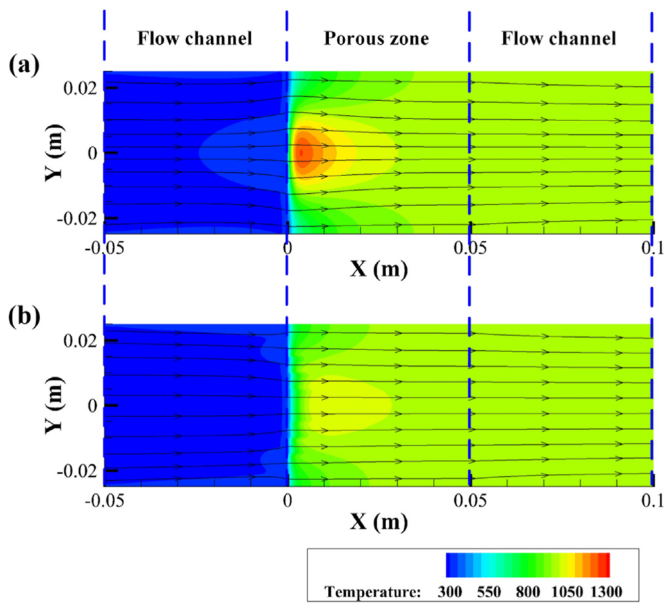

Fig. 21. Temperature distribution in the porous volumetric solar receiver (a) Uniform porous media; (b)Radial graded porous media (Adapted with permission [164]. Copyright 2020, Elsevier).

materials include intrinsic materials, semiconductor-metal tandems, multilayer structures, cermets, surface texturing, and photonic crystals based designs [180]. Considering for the high working temperature in the CSP system, the spectrally selective receiver mainly focuses on the application of intrinsic materials or multilayer structures.

Olalde et al. [181] designed two spectrally selective porous volumetric solar receivers that both had gradually-varied radiative properties. One receiver was a honeycomb structure with transparent matrix, and the absorptivity was gradually increased by controlling the deposit thickness of chemical vapor deposition. The other receiver was constructed by a packed bed which was a mixed composition of particles with different radiative properties. The front layer was designed to be transparent in the solar spectrum while the absorptivity was progressively increased by adding more absorptive particles in the rear. The thermal radiative loss was limited because the transparent matrix had a larger absorptivity in the infrared spectrum. Pitz-Paal et al. [182] proposed a double-layer porous volumetric solar receiver with quartz glass in the front of the SiSiC ceramic as shown in Fig. 22(a). The quartz glass was transparent to solar radiation and the SiSiC ceramic can volumetrically absorbed solar energy. The thermal radiation emitted by the high temperature ceramic was reabsorbed by the quartz glass because it could effectively absorb the infrared emission. It was demonstrated that the thermal efficiency could be improved from 68.5% to 75.0% using this double-layer design when the outlet air temperature reached above 900 °C.

Similarly, Menigault et al. [183] also developed double-layer spectrally selective porous volumetric solar receivers. The solar radiation absorption layer was placed in the rear which was composed of SiC particles as shown in Fig. 22(b). The front layer consisted of glass spheres or pure silica honeycomb structures, which can absorb the thermal emission from the SiC particles. For the receiver with glass spheres, the outlet air temperature was lower than 500 °C due to the temperature limitation of glass. However, by fitting the experimental data from the high temperature zone, it was shown that the double-layer receiver outperformed when the outlet air temperature reached above 600 °C. For the double-layer receiver with silica honeycomb structures, the outlet air temperature can achieve 600 °C, and the thermal efficiency was improved from 71% to 76%, compared with the pure SiC porous volumetric solar receiver. Sedighi et al. [184] further designed and investigated the packed bed volumetric solar receiver with spheres of different transmittance. In order to improve the penetration depth and achieve the volumetric effect, the quartz spheres with high transmittance were installed in the front of the receiver. Meanwhile, the ceramic spheres which have large absorptance were located in the rear of the receiver to prevent the transmission loss. The outlet temperature of the variable transmittance quartz packed bed was significantly increased compared with the quartz and ceramic packed beds, and a high index of volumetric effect of 1.45 was achieved.

Moreover, investigations on the application of intrinsic spectrally selective materials for porous volumetric solar receivers have been carried out. Zhu and Xuan [185] numerically investigated a porous volumetric porous solar receiver with a $MoSi_2$-Si_3N_4 spectrally selective coating. This coating was compared with the ideal coating material with a cut-off wavelength of 2 μm. Compared with the regular receivers, the receivers with the $MoSi_2$-Si_3N_4 coating and the ideal coating could improve the thermal efficiency by 1.2% and 2.9%, respectively. Some studies focused on the radiative properties of ultra-high temperature ceramics such as HfB_2 and TaC [186,187], but few studies applied these ceramics to porous volumetric solar receivers and investigated the performance improvement. Rousseau et al. [188] utilized the pore-scale reconstruction and ray tracing methods to analyzed the emissivity of the porous skeleton with TaC, ZrC, and ZrB_2 coatings. Compared with SiC which has a large emissivity in the infrared spectrum, the ZrB_2 coating can largely achieve the spectrally selective absorption, and the emissivity in the infrared spectrum and absorptivity in the solar spectrum were 0.31 and 0.81, respectively. Recently, Du et al. [189] proposed the porous volumetric solar receiver based on molten salt as a heat transfer fluid, and the Hitec and air receivers were compared based on the direct pore-scale numerical simulation method. By utilizing the intrinsic spectrally selective absorptivity, the molten salt could be regarded as the transparent window to solar radiation which absorbs the thermal radiation emitted by the porous structure. The results showed that the thermal efficiency of the Hitec porous volumetric solar receiver could be improved by 9.6% at receiver's outlet temperature of 1000 K as compared with the air porous volumetric solar receiver.

5. Conclusions and outlooks

Since the 1980s, the experimental studies have accumulated experience in material selection, structural design, and system operation of porous volumetric solar receivers. The ceramic materials have become a better choice due to the higher upper temperature limit and oxidation resistance. Silicon carbide has been widely used due to their good absorption performance in the solar spectrum and high temperature resistance characteristics. For the atmospheric porous volumetric solar receiver, Solar Tower Jülich demonstrated the mature design which utilized the modular design concept and air return system. Such design could be easily scale-up and the overall thermal efficiency was high. For the pressurized porous volumetric solar receiver, problems exist for the secondary concentrator and transparent window. The non-uniform and

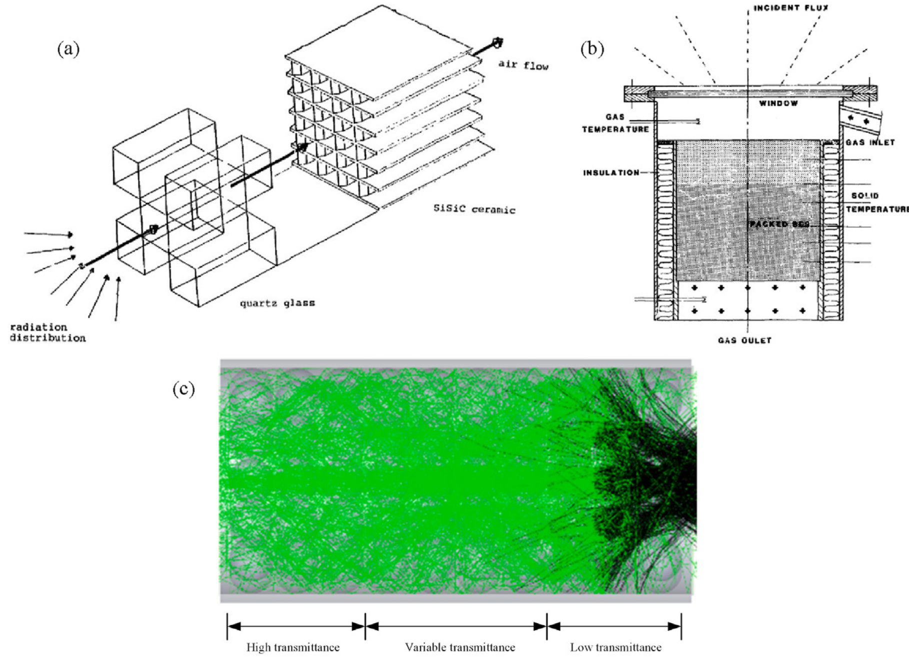

Fig. 22. Double-layer porous volumetric solar receivers with spectral selective absorption (a)Pitz-Paal et al. (Reproduced with permission [182]. Copyright 1991, Elsevier); (b)Menigault et al. (Reproduced with permission [183]. Copyright 1991, Elsevier); (c)Reproduced from Sedighi et al. (Adapted with permission [184]. Copyright 2020, Elsevier).

high temperature working condition severely influence the thermal efficiency and long-term usage of the entire system. Hence, it is crucial to test the long-term stability of the receiver, and establish a large-scale CSP system based on a porous volumetric solar receiver to assess its compatibility with various power cycles.

Due to the complex structure of the porous media, limited experimental measurement conditions, and high experimental costs, numerical simulation methods have become an important means for analyzing the fluid flow and heat transfer performance of porous volumetric solar receivers. The direct pore-scale numerical simulation method could restore the detailed structure of the porous media and accurately reveal the fluid flow and heat transfer processes inside the receiver. However, this method requires a lot of computational resources and a full-scale numerical model considering the solar radiation transfer and coupled heat transfer is difficult to build. On the other hand, the volume averaging simulation method can realize the rapid evaluation of the thermal performance of porous volumetric solar receivers which becomes the main tool to conduct parametric sensitivity analysis and optimization design. However, the accuracy of the simulation results of this method depends on the rational selection of the empirical parameters. Therefore, it is necessary to establish a full-scale direct pore-scale model to investigate the morphologies of the porous skeleton and the transient performance of the porous volumetric solar receiver. At the REV scale, a more efficient optimization algorithm can be integrated into the volume-averaging simulation method to consider irregular shapes of porous structures and complex air flow paths, thereby optimizing the overall performance of the porous volumetric solar receivers.

In order to enhance the volumetric absorption of the porous media, many studies have focused on the geometrical parameters optimization especially along the axial direction. However, different studies have not achieved consistent results in the thermal efficiency improvement of the multilayer design. Furthermore, although the 3D printing technique has been utilized for fabricating porous media with non-uniform structures, the experimental validation of the design concept remains incomplete. It is essential to propose a general evaluation index and summarize an optimization roadmap. On the other hand, many numerical and experimental studies have proved the high efficiency of the spectrally selective absorption concept. But experimental work is still needed for the accurate measurements of radiative properties of the porous media and heat transfer fluid. Further experiments are necessary to verify the excellent thermal performance of the spectrally selective porous volumetric solar receiver.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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